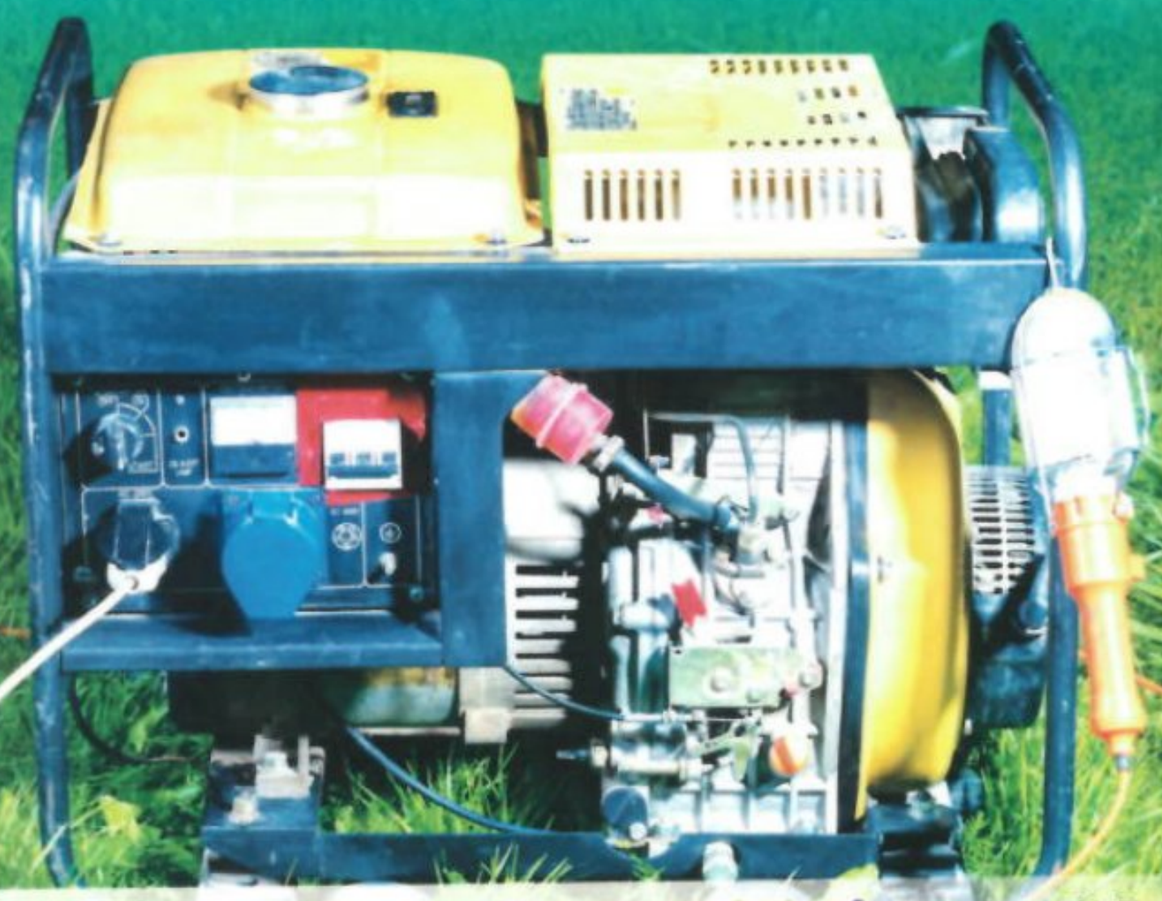
Chapter 3

Questions on Lesson Three

Electric Generator (Dynamo)

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply ⁂ Higher Order Thinking Skills

Important guidelines

$(emf)_{inst} = (emf)_{max} \sin\theta$
$= NBA\omega \sin\theta$
$= NBA\omega \sin \omega t$
$= NBA \times 2\pi f \sin 2\pi ft$

Instantaneous

$(emf)_{eff} = \dfrac{(emf)_{max}}{\sqrt{2}}$
$= 0.707\ (emf)_{max}$

Effective

$(emf)_{max} = NBA\omega$

Where:

$\omega = \dfrac{\theta}{t} = 2\pi f = \dfrac{v}{r}$, $f = \dfrac{1}{T}$

Maximum

The induced emf generated in the AC dynamo

Average

If the coil rotates from the perpendicular position

90° ($\frac{1}{4}$ cycle)	180° ($\frac{1}{2}$ cycle)	270° ($\frac{3}{4}$ cycle)	360° (complete cycle)
$\Delta\phi_m = -BA$	$\Delta\phi_m = -2BA$	$\Delta\phi_m = -BA$	$\Delta\phi_m = 0$
$(emf)_{av} = NBA \times 4f$	$(emf)_{av} = NBA \times 4f$	$(emf)_{av} = NBA \times \frac{4}{3}f$	$(emf)_{av} = 0$

If the coil rotates from the parallel position

90° ($\frac{1}{4}$ cycle)	180° ($\frac{1}{2}$ cycle)	270° ($\frac{3}{4}$ cycle)	360° (complete cycle)
$\Delta\phi_m = BA$	$\Delta\phi_m = 0$	$\Delta\phi_m = -BA$	$\Delta\phi_m = 0$
$(emf)_{av} = -NBA \times 4f$	$(emf)_{av} = 0$	$(emf)_{av} = NBA \times \frac{4}{3}f$	$(emf)_{av} = 0$

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Maximum emf $(emf)_{max} = NBA\omega$

- $\times \sin \theta$ → Instantaneous emf
- $\times \frac{1}{\sqrt{2}}$ → Effective emf
- $\times \frac{2}{\pi}$ → Average emf through $\frac{1}{4}$ cycle **Or** through $\frac{1}{2}$ cycle from the perpendicular position to the field
- $\times \frac{2}{3\pi}$ → Average emf through $\frac{3}{4}$ cycle

- **To determine the instantaneous value of alternating current $(I)_{inst}$:**

  $(I)_{inst} = I_{max} \sin \theta = I_{max} \sin 2\pi f$

  (Where: I_{max} is the maximum value of the alternating current)
- **To determine the effective value of alternating current $(I)_{eff}$:**

  $(I)_{eff} = \frac{(I)_{max}}{\sqrt{2}} = 0.707\ (I)_{max}$
- **The number of times the AC current reaches maximum value per second (starting from zero position)** $= 2f$
- **The number of times the AC current reaches zero per second (starting from zero position)** $= 2f + 1$

First Multiple choice questions

1. The direction of the electric current produced in the dynamo coil can be determined using ..... rule.
   (a) Fleming's left hand (b) Ampere's right hand
   (c) Fleming's right hand (d) right hand screw

2. The rate with which the coil intercepts the lines of the magnetic field in the dynamo is maximum when the plane of the coil is ..... to the flux lines.
   (a) perpendicular (b) parallel
   (c) inclined with 30° (d) inclined with 60°

3. The time rate of cutting the magnetic flux lines by the dynamo's coil during its rotation becomes zero when the plane of the coil becomes ..... .
   (a) inclined to the field by an angle of 45° (b) parallel to the field
   (c) perpendicular to the field (d) inclined to the field by an angle of 30°

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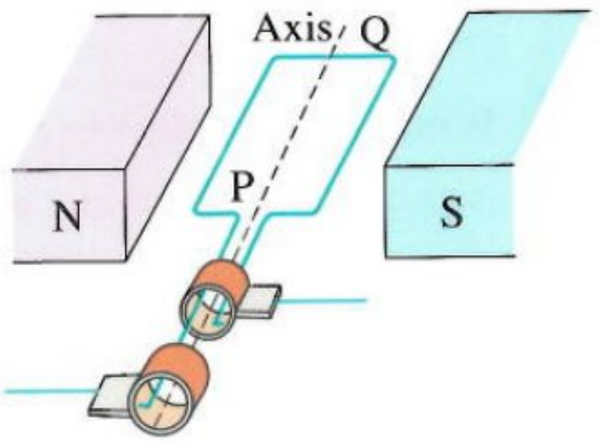
4 At the instant when the AC dynamo coil becomes parallel to the direction of the magnetic flux, the magnetic flux (ϕ_m) through the coil and the induced emf in the coil become ..... .

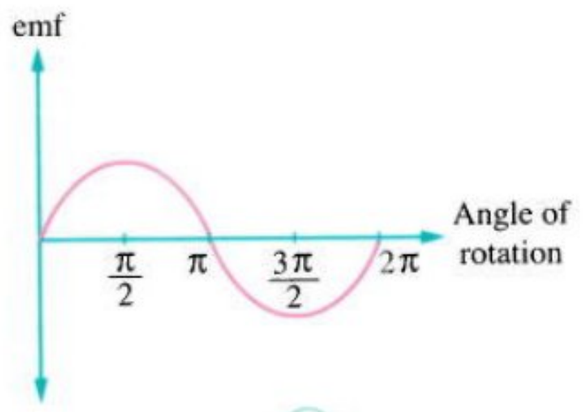
	ϕ_m	emf
a	maximum value	zero
b	zero	maximum value
c	maximum value	maximum value
d	zero	zero

5 The induced electromotive force which is obtained from an AC dynamo reaches its maximum value when the plane of the coil with respect to the magnetic field becomes ..... .

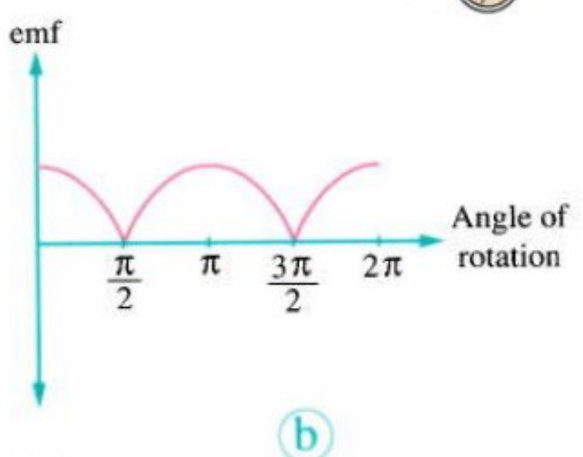
a perpendicular
b parallel
c inclined by angle of 45°
d inclined by angle of 60°

6 A rectangular coil is rotating about its axis PQ between the poles of a magnet, starting from the position shown in the opposite figure, which of the following graphs represents the change in the induced emf in the coil through one complete rotation?

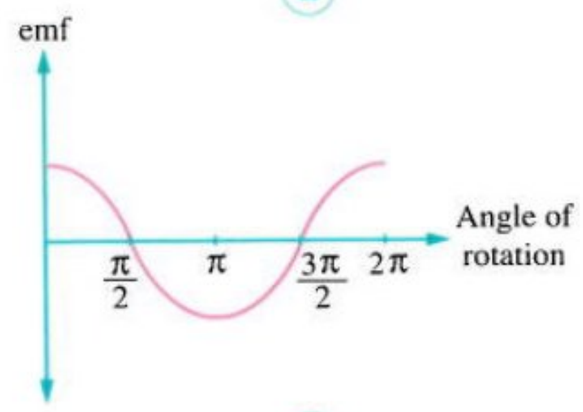




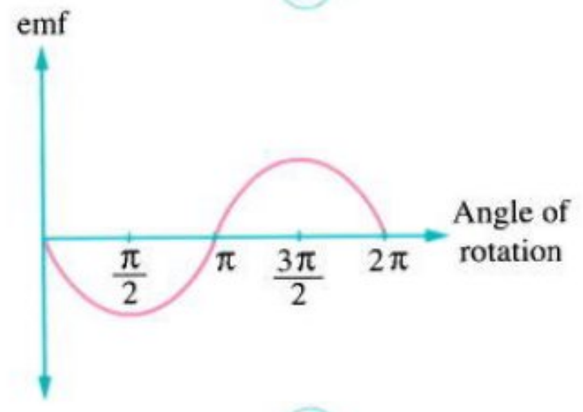
a



b



c



d

7 An AC dynamo coil rotates in a uniform magnetic field at an angular speed ω, so the periodic time of the coil equals ..... .

a $\frac{\omega}{\pi}$
b $\frac{2\pi}{\omega}$
c $\frac{\pi}{\omega}$
d $\frac{\omega}{2}$

8 An AC dynamo coil rotates about its longitudinal axis at an angular speed of 377.2 rad/s, then the frequency of the current equals ..... .

a 50 Hz
b 60 Hz
c 75 Hz
d 100 Hz

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9 If the number of turns of the dynamo's coil is doubled and its angular speed (ω) is decreased to its quarter, so the maximum generated emf from that coil ..... .

(a) increases to double (b) decreases to its half
(c) remains constant (d) decreases to its quarter

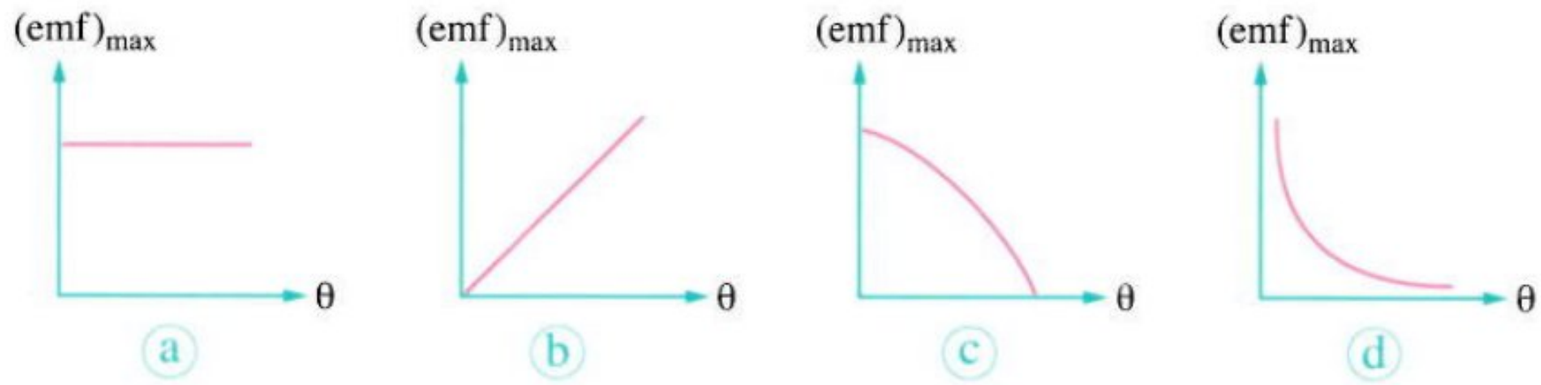
10 An electric generator's coil rotates at an angular speed of 281 rad/s producing an induced emf of a maximum value 120 V, so the angular speed required for the same coil to produce an induced emf of a maximum value that equals 480 V is ..... .

(a) 2.7 rad/s (b) 70.3 rad/s (c) 205 rad/s (d) 1124 rad/s

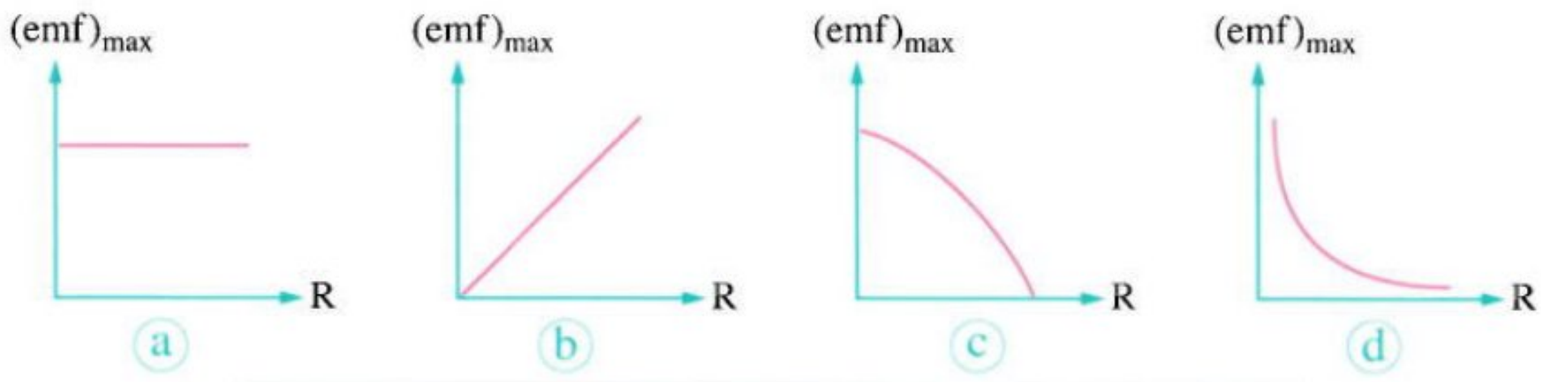
11 If the generated $(emf)_{max}$ in a dynamo's coil was 100 V when the frequency of the rotation of dynamo's coil was f and when the frequency increased by 25 Hz, the $(emf)_{max}$ increased to 150 V, so the value of frequency f equals ..... .

(a) 25 Hz (b) 50 Hz (c) 100 Hz (d) 150 Hz

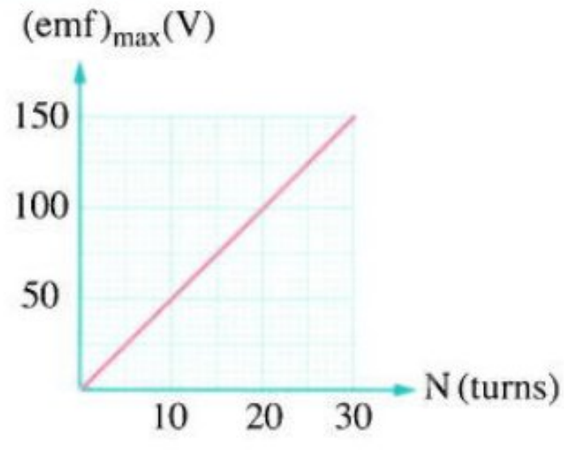
12 Which of the following graphs represents the relation between the magnitude of the maximum induced electromotive force $(emf)_{max}$ generated in the coil of a dynamo and the angle of rotation (θ) of the coil?



13 Which of the following graphs represents the relation between the magnitude of the maximum induced electromotive force $(emf)_{max}$ produced in the coil of a dynamo and the resistance of the dynamo's coil (R)?



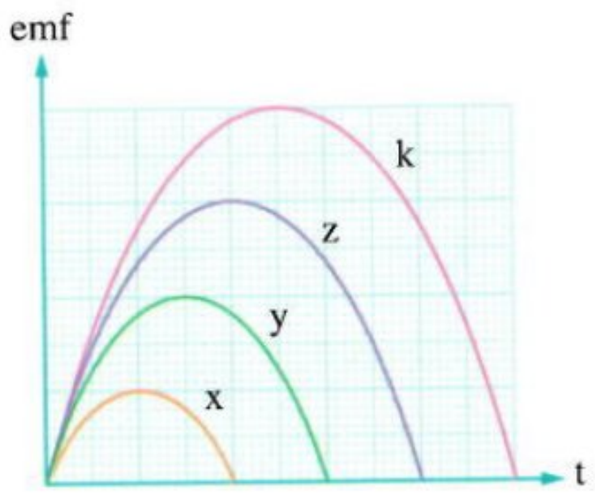
14 The opposite graph represents the variation of the maximum induced electromotive force $(emf)_{max}$ in a dynamo's coil that has a changeable number of turns with respect to the number of turns of the coil (N). If the angular speed of rotation of the coil is 100 rad/s and the magnetic flux density is 0.4 T, then the cross-sectional area of the coil equals ..... .



(a) 0.5 m^2 (b) 0.2 m^2 (c) 0.25 m^2 (d) 0.125 m^2

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15 ✱ Four electric generators x, y, z, k which have the same number of turns and are affected by similar magnetic fields. The opposite graph represents the relation between the induced (emf) in the coil of each of them through half cycle for each coil and the time (t), so the relation between the face areas of these coils is .....

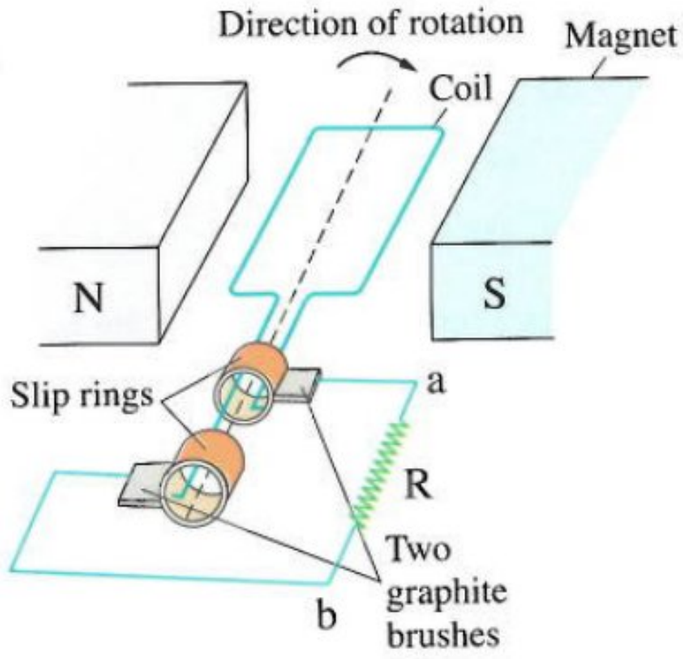


(a) $A_x > A_y > A_z > A_k$ (b) $A_x = A_y = A_z = A_k$

(c) $A_k > A_y > A_z > A_x$ (d) $A_k > A_z > A_y > A_x$

16 ✱ The opposite figure shows an electric dynamo that has a coil of area 0.5 m^2 and number of turns 400 turns. The coil rotates about its axis of rotation by an angular speed of 10 π rad/s where the maximum value of the induced emf in the coil is 628 V, so ..... .

(Where: $\pi = 3.14$)



	The magnetic flux density acting on the coil	The direction of the induced current through the external circuit at the shown instant
(a)	0.01 T	from a to b
(b)	0.01 T	from b to a
(c)	0.1 T	from a to b
(d)	0.1 T	from b to a

17 The instantaneous induced emf in a dynamo's coil is determined from the relation; $(emf)_{inst} = NBA\,\omega \sin\theta$, which of the following statements **does not** correctly describe angle θ in this relationship?

(a) Angle θ is the angle between the perpendicular to the direction of the magnetic field and the plane of the coil

(b) Angle θ is the angle between the direction of the magnetic field and the normal to the coil's plane

(c) Angle θ is the angle between the direction of the magnetic field and the direction of velocity of one side of the coil

(d) Angle θ is the angle between the direction of the magnetic field and the coil's plane

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18 The induced instantaneous emf in the coil of the AC dynamo when the angle between the plane of the dynamo coil and the direction of the magnetic flux is:

(i) 60°, is ..... .

(a) $\frac{\sqrt{3}}{2}$ of the maximum value (b) $\frac{1}{2}$ of the maximum value

(c) equal to the maximum value (d) equal to the effective value

(ii) 45° is ..... .

(a) $\sqrt{2}$ of the maximum value (b) half the maximum value

(c) equal to the maximum value (d) equal to the effective value

19 ✱ If the maximum induced emf in the coil of a dynamo is 200 V, so the value of the instantaneous induced emf when:

(i) The coil reaches $\frac{1}{12}$ of one cycle since the moment at which the emf was equal to zero is ..... .

(a) 50 V (b) 100 V (c) 150 V (d) 200 V

(ii) The angle between the normal to the plane of the coil and the flux lines is 30° equals ..... .

(a) 50 V (b) 100 V (c) 150 V (d) 200 V

(iii) The plane of the coil becomes inclined by an angle 60° on the field is ..... .

(a) 100 V (b) 70 V (c) 50 V (d) 20 V

20 ✱ AC dynamo, its coil is composed of 200 turns, the cross-sectional area of each is 6×10^{-2} m^2, rotates in a magnetic field with a speed of 1800 cycles per minute. If the magnetic flux density was 0.1 T, then:

(i) The induced emf in the coil when passing in each of the following positions:

① The plane of the coil is perpendicular to the field equals ..... .

(a) 0 (b) 72 V (c) 144 V (d) 216 V

② The plane of the coil is parallel to the field equals ..... .

(a) 0 (b) 226.29 V (c) 452.58 V (d) 678.87 V

③ The plane of the coil is inclined by an angle of 60° on the direction of the field equals ..... .

(a) 113.15 V (b) 144.5 V (c) 155.3 V (d) 169.2 V

(ii) The average induced emf in the following cases:

① through a quarter cycle from the perpendicular position to the field equals ..... .

(a) 36 V (b) 72 V (c) 144 V (d) 288 V

② through half cycle from the perpendicular position to the field equals ..... .

(a) 0 (b) 144 V (c) 288 V (d) 300 V

③ through complete cycle starting from zero position equals ..... .

(a) 576 V (b) 144 V (c) 36 V (d) 0

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21 ✱ An AC dynamo coil of dimensions 5 cm, 10 cm that is composed of 420 turns is placed in a uniform magnetic field of flux density 0.4 tesla such that the plane of the coil is perpendicular to that field. If the coil rotates with a rate of 1000 cycles per minute, then the induced emf in each of the following positions:

(i) After $\frac{1}{4}$ cycle from the first position equals ..... .

(a) 22 V (b) 44 V (c) 66 V (d) 88 V

(ii) After 150° from the first position equals ..... .

(a) 22 V (b) 44 V (c) 66 V (d) 88 V

22 ✱ A dynamo coil of cross-sectional area 4×10^{-2} m^2 that consists of 70 turns rotates with a speed of 3600 cycles/minute in a magnetic field of flux density 0.5 tesla. If it started motion from the instant when its plane was perpendicular to the direction of the field, after $\frac{1}{720}$ s from that instant, emf will equal ..... .

(a) 132 V (b) 264 V (c) 276.6 V (d) 342.8 V

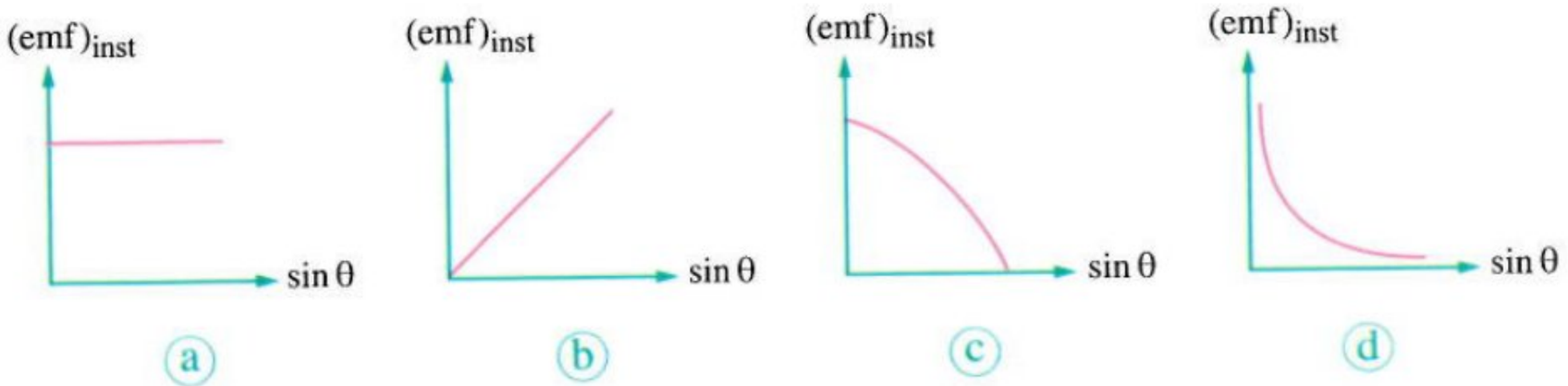
23 A coil of an AC dynamo consists of 600 turns, the area of each of them is 25 cm^2, if the coil is rotated about an axis perpendicular to a uniform magnetic flux of density B with a constant angular speed (ω), an induced emf is generated where it is given by the relation; emf = 12.5 sin (100 πt), then the magnetic flux density (B) is approximately ..... .

(a) 2.7×10^{-6} T (b) 2.7×10^{-4} T (c) 2.7×10^{-2} T (d) 2.7 T

24 ✱ If the induced electromotive force in the coil of an AC dynamo of 100 turns is given by the relation; emf = 100 π sin (100 πt), then the maximum value of the magnetic flux that passes through each turn of the coil during its rotation is ..... .

(a) 10^{-2} Wb (b) 2×10^{-3} Wb (c) 2×10^{-4} Wb (d) 10^{-4} Wb

25 Which of the following graphs represents the relation between the magnitude of the instantaneous electromotive force $(emf)_{inst}$ generated in a dynamo's coil and the sine of the angle of rotation of the coil (sin θ) when the coil starts the rotation from zero position?



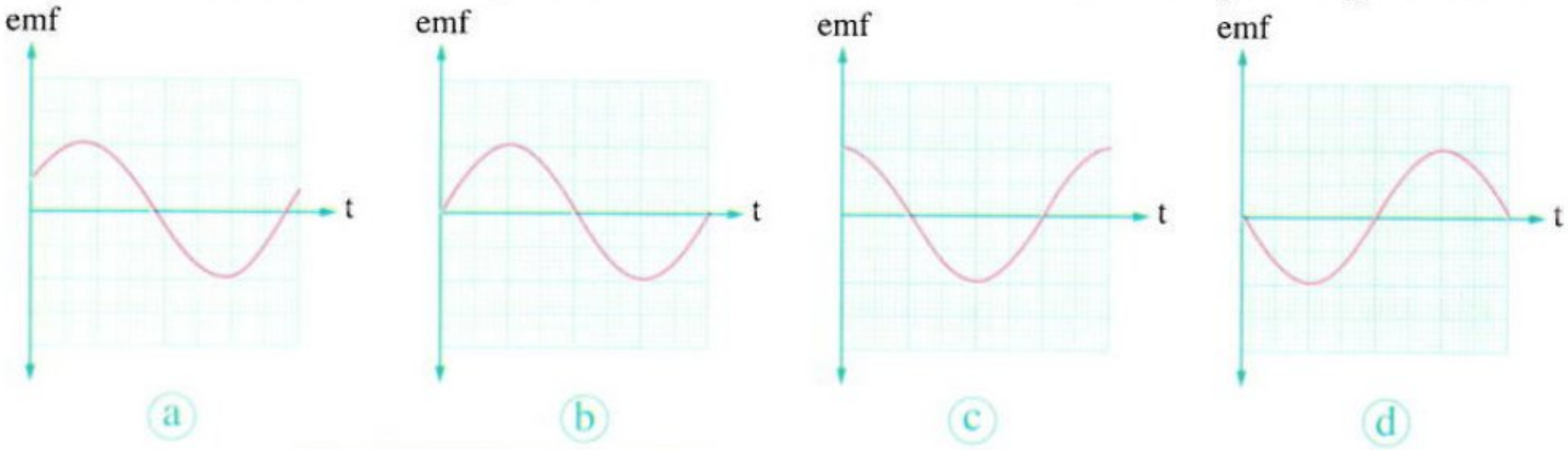
(a) (b) (c) (d)

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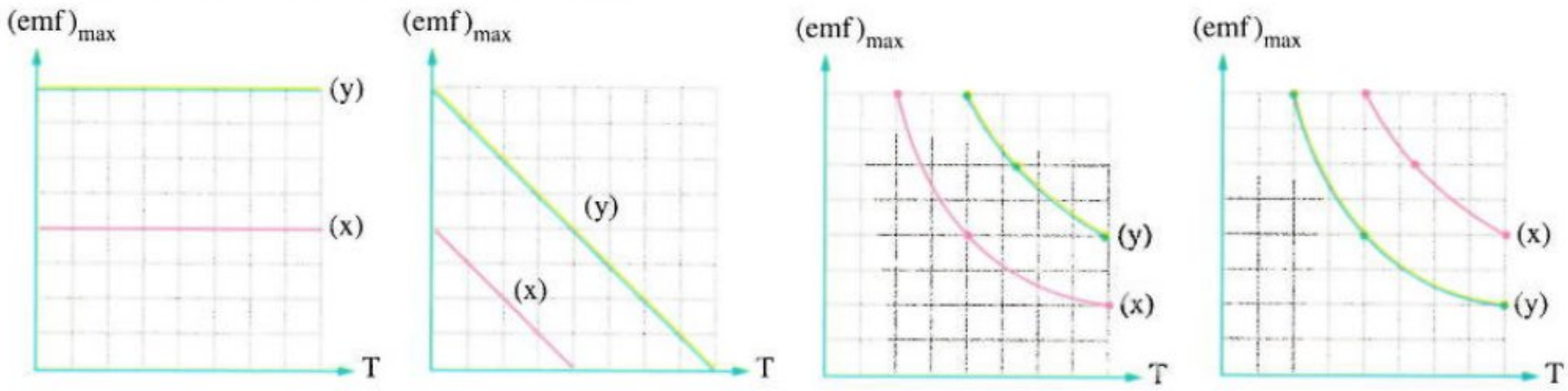
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26 Which of the following graphs represents the relation between the instantaneous induced electromotive force and the time in a simple dynamo coil when the coil starts the rotation from the position where the plane of the coil was inclined to the field by an angle of 60°?



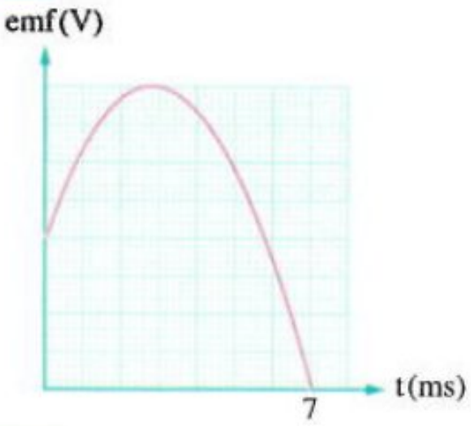
(a) (b) (c) (d)

27 Two dynamo coils X, Y, their number of turns are N, 2 N respectively are rotating in two magnetic fields having the same flux density. If the frequency of rotation of the two coils is changed, which of the following graphs represent relation between the maximum value of the induced electromotive force (emf_{max}) in each of the two coils and the periodic time (T) for the rotation of each of them?



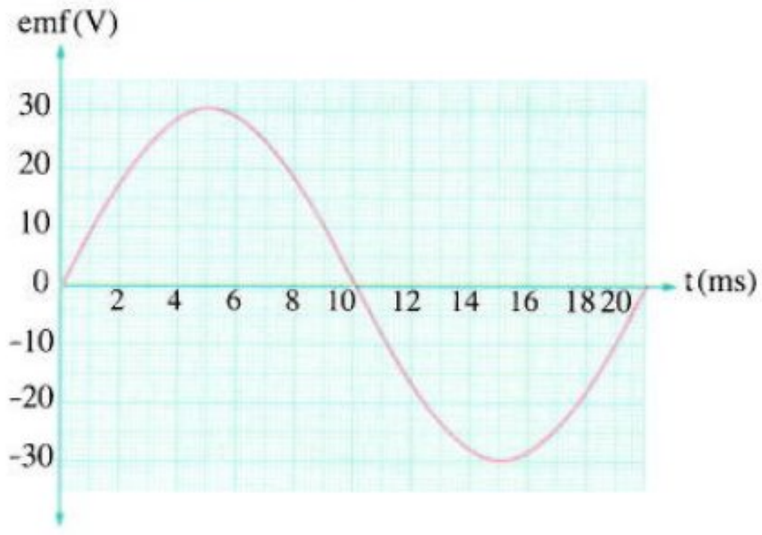
(a) (b) (c) (d)

28 The opposite graph represents the relation between the induced emf in an AC dynamo coil and time (t) within 7 ms, so the rotation frequency of the dynamo coil equals ..... .



(a) 59.5 Hz (b) 71.4 Hz
(c) 119 Hz (d) 142.8 Hz

29 The opposite graph represents the relation between the induced emf through a complete cycle and the time (t) for an AC dynamo that consists of a coil of face area 0.125 m^2 and number of turns 200 turns. Thus;



(i) The magnetic flux density in which the coil of the dynamo is rotating equals ..... .

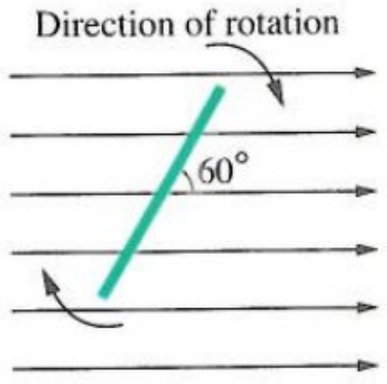
(a) 1.4×10^{-3} T (b) 2.6×10^{-3} T (c) 3.8×10^{-3} T (d) 4.2×10^{-3} T

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(ii) When the plane of the coil makes angle of 60° with the direction of the magnetic field, the instantaneous generated emf equals ..... .

(a) 12 V (b) 14 V (c) 15 V (d) 18 V

30 An AC dynamo coil rotates with a frequency of 120 Hz, if the coil at a certain moment makes an angle 60° with the magnetic flux, so from that moment, the instantaneous induced emf in the coil vanishes after a time of ..... .



(a) 0.12 ms

(b) 8.33 ms

(c) 6.33 ms

(d) 3.47 ms

31 The opposite two figures represent two models for the coils of an AC dynamo (1) and (2) of number of turns 10 and 20 turns respectively. The flux density that acts on each of the two coils is B and each of them rotates such that the linear speed of the side that is parallel to the axis of rotation is v, then the ratio $\left(\frac{(emf_{max})_1}{(emf_{max})_2}\right)$ is ..... .

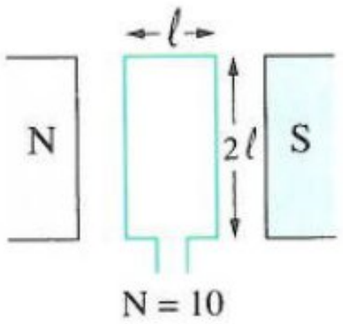


Figure (1)

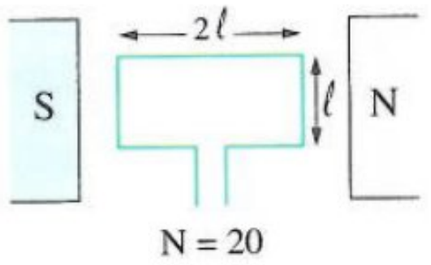
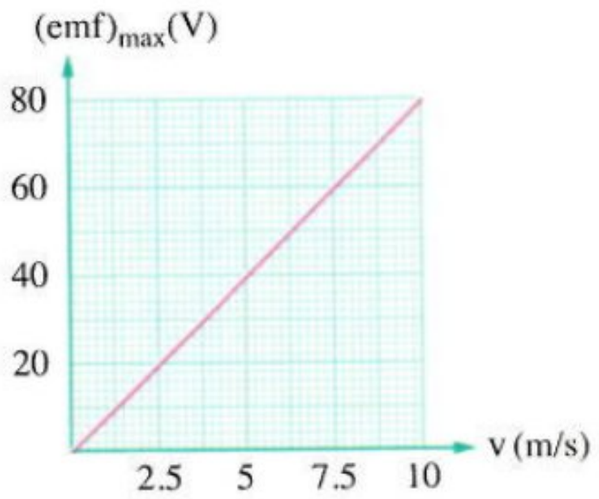


Figure (2)

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{1}$ (d) $\frac{2}{1}$

32 An AC dynamo coil consists of 200 turns where the length of each is 10 cm and rotates with a uniform speed about an axis parallel to its length. The opposite graph represents the relation between the maximum induced electromotive force $(emf)_{max}$ in the dynamo coil and the linear speed for one of the longitudinal lines, so the magnetic flux density acting on the coil equals ..... .

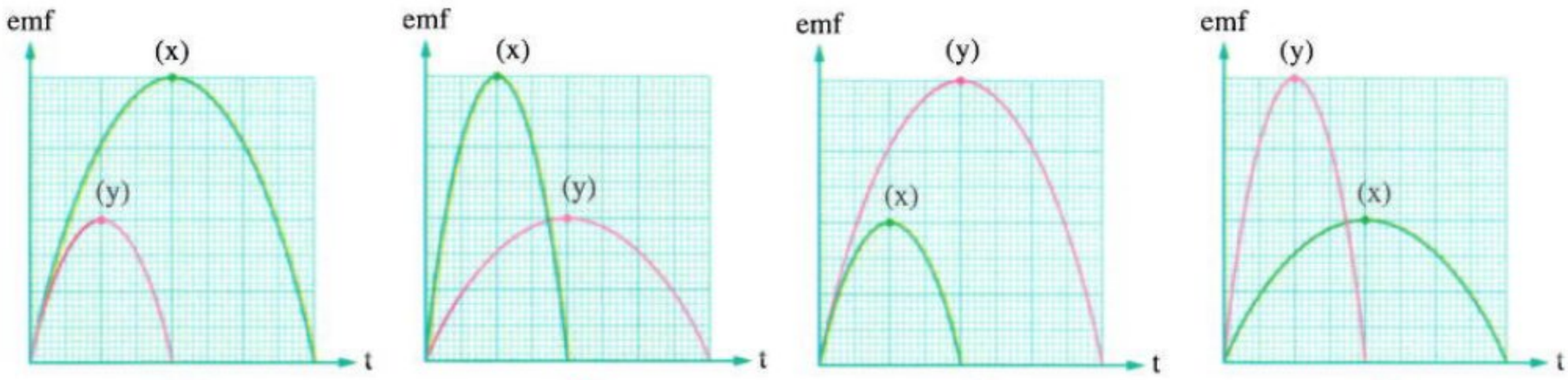


(a) 0.1 T (b) 0.2 T (c) 0.3 T (d) 0.4 T

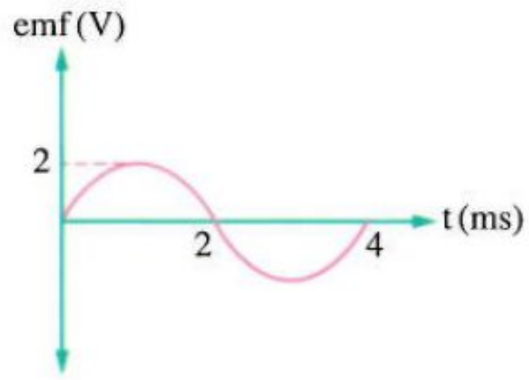
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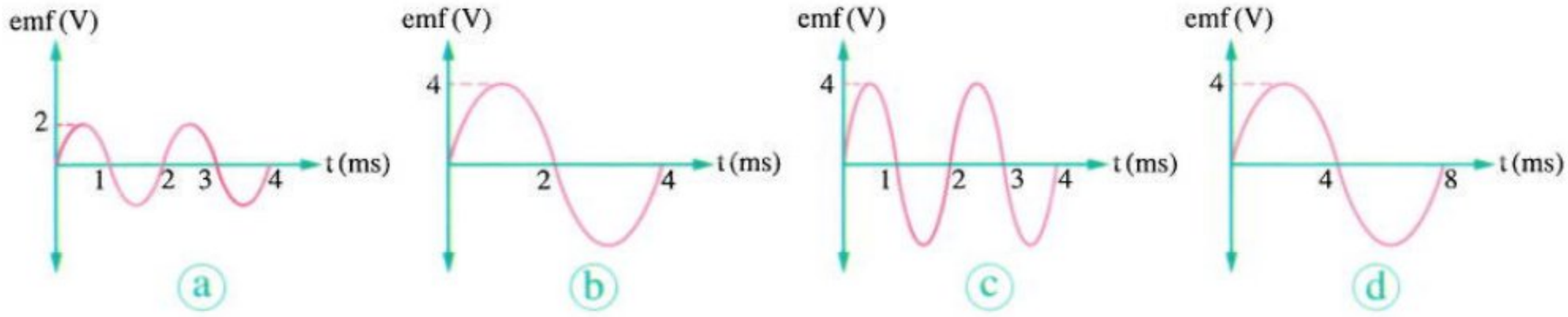
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33 Two AC dynamos (x), (y), having the same number of turns but their lengths are ℓ, 2ℓ and their widths are w, $\frac{1}{2}$w respectively, each of them is rotating about an axis parallel to its length with the same linear speed inside two magnetic fields of the same flux density, so which of the following graphs represents the relation between the induced emf in each of the coils and time (t) through half cycle?

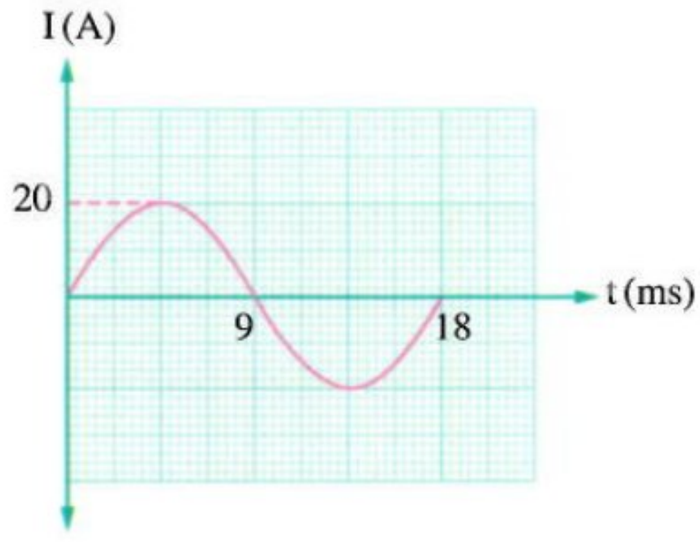


34 The opposite graph expresses the relation between the instantaneous induced emf in a dynamo coil of rotation frequency f and time (t), so if the rotation frequency of the coil is increased to 2 f, then the graph which is expressing the same relation would be ..... .





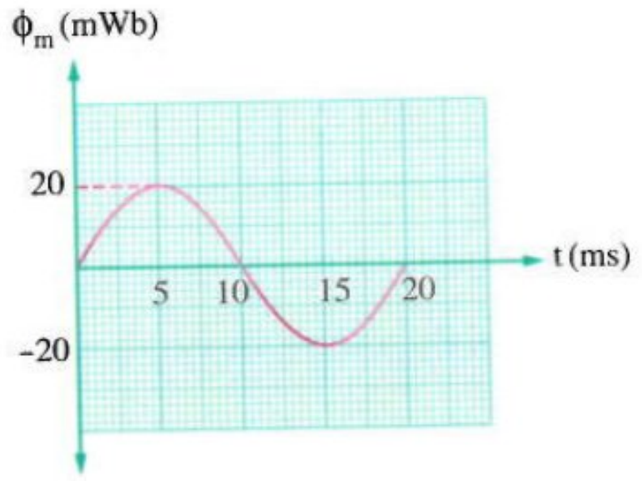
35 ✱ The opposite graph represents the relation between the induced electric current intensity in the coil of an AC dynamo and the time of rotation of its coil. If the resistance of the dynamo's circuit is 16.5 Ω, then the instantaneous induced electromotive force that is generated after 12 ms from the zero position is approximately equal to ..... .



(a) 165 V (b) – 176 V (c) 219 V (d) – 286 V

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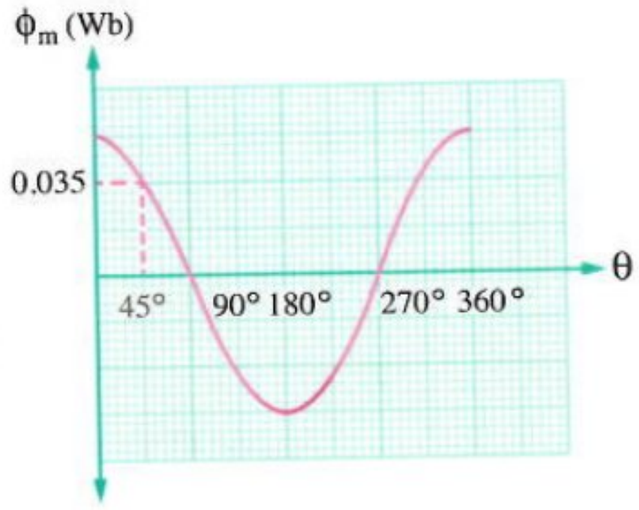
36 An AC dynamo's coil consists of 70 turns, where the opposite graph represents the relation between the magnetic flux (ϕ_m) which passes through the coil and time. So, value of the induced electromotive force in the coil at (t = 10 ms) equals ..... .



(a) 0 (b) 220 V

(c) 440 V (d) 880 V

37 ✳ The opposite graph shows the change of the magnetic flux (ϕ_m) through a complete cycle for the coil of a dynamo that consists of 8 turns and its frequency of rotation is 50 Hz, then:

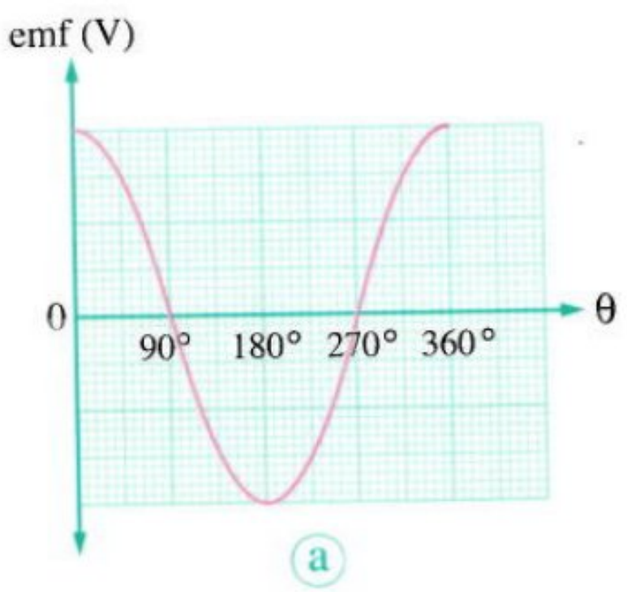


(i) The magnitude of the induced emf generated in the coil after quarter of the periodic time equals ..... .

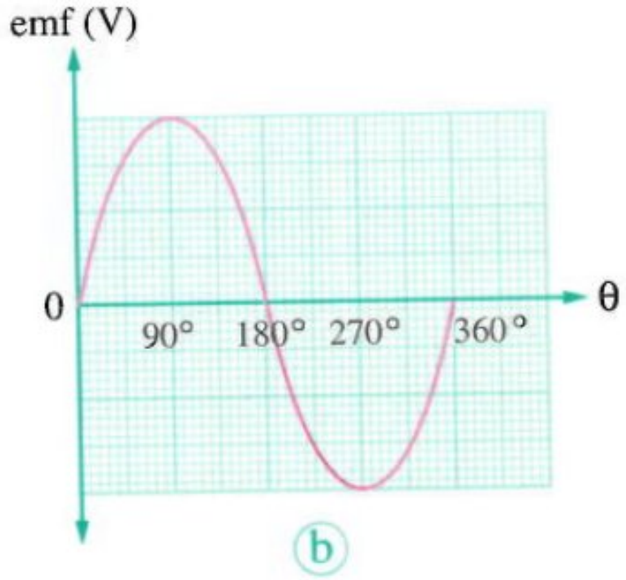
(a) 123.2 V (b) 100 V

(c) 61.6 V (d) 0

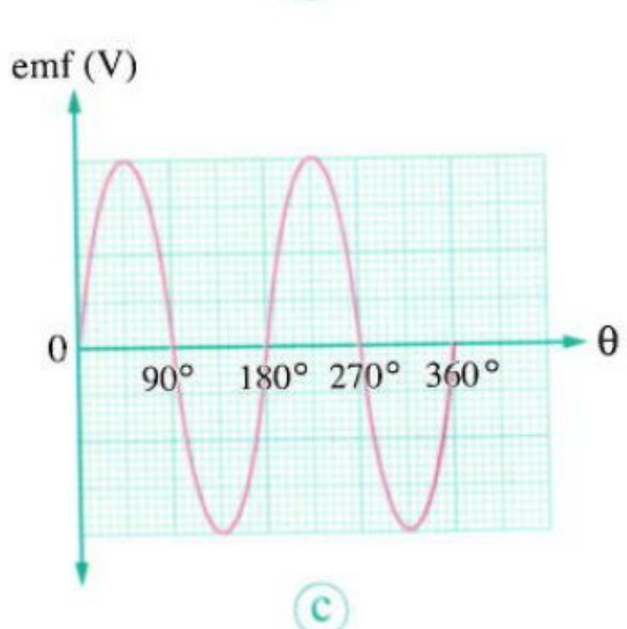
(ii) Which of the following choices represents the relation between the induced emf generated in the coil of the dynamo and the angle (θ) through a complete cycle?



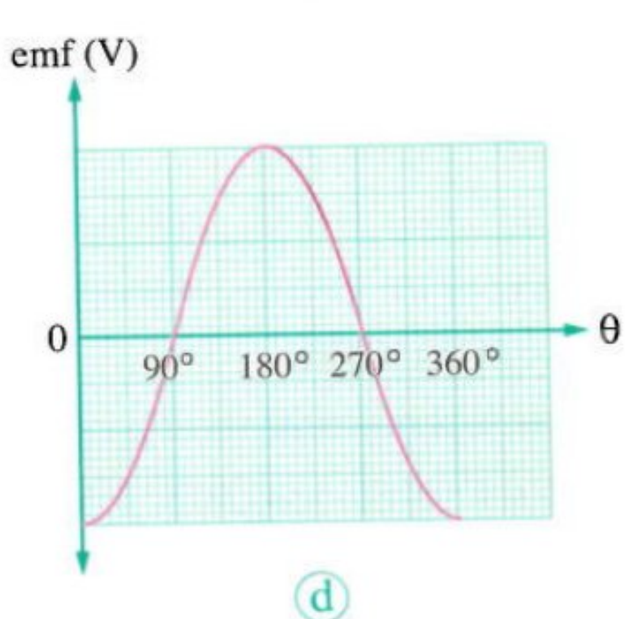
(a)



(b)



(c)



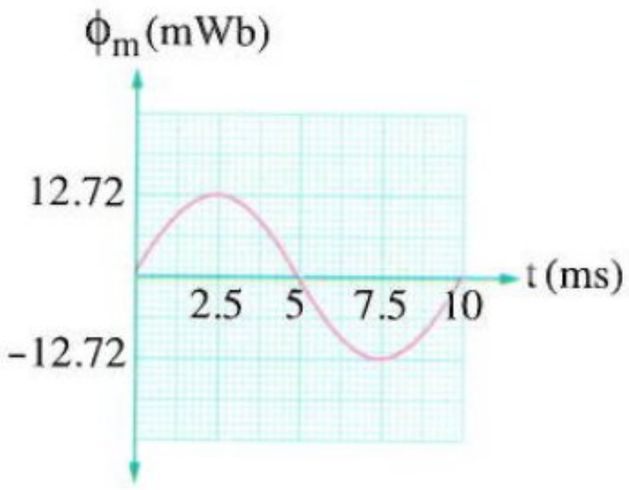
(d)

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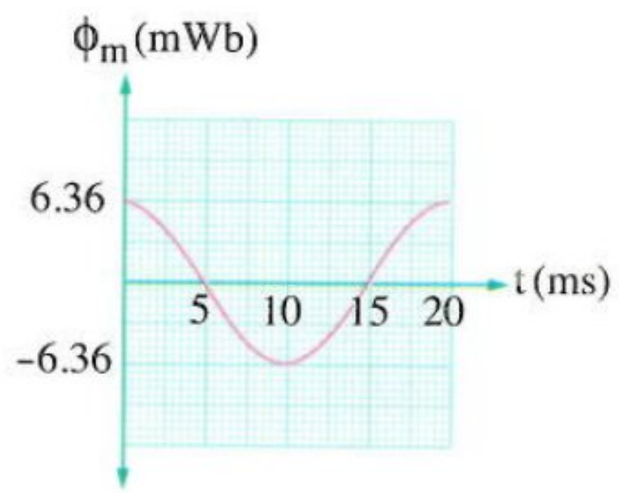
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38 An AC dynamo coil of resistance 5 Ω consists of 100 turns and connected to an external circuit of resistance 5 Ω where the current passing through it at any moment (t) is given by the relation ; $I = 20 \sin (100 \pi t)$.

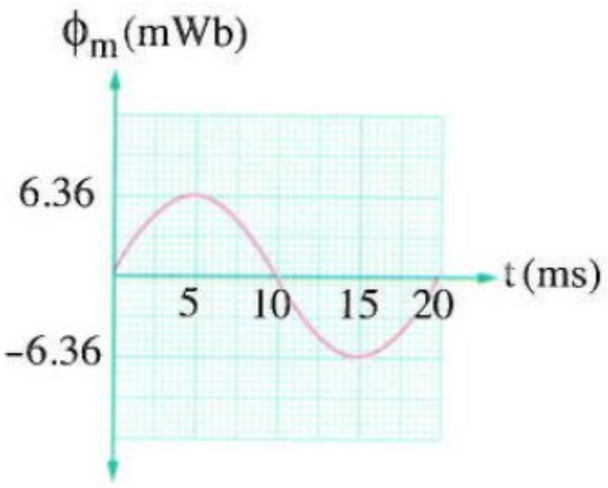
Which of the following graphs represents the relation between the magnetic flux (ϕ_m) which passes through the coil and time (t)?



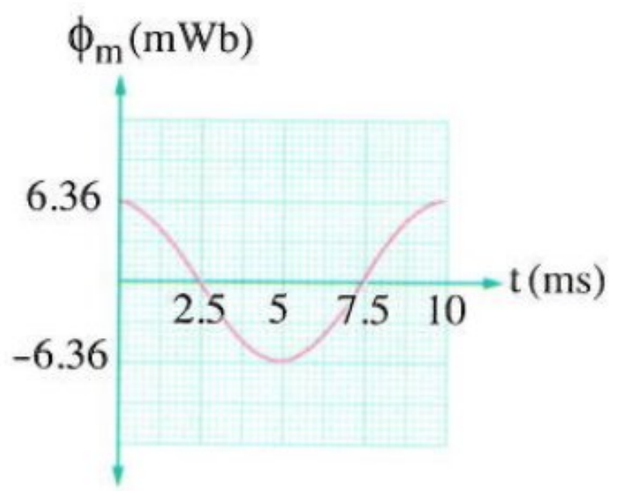
a



b



c



d

39 The average induced emf for a complete revolution of an AC dynamo equals ..... .

a emf_{eff} b emf_{inst} c emf_{max} d zero

40 When a coil rotates in a magnetic field starting from zero position, the direction of the generated induced emf changes every ..... cycle.

a $\frac{1}{4}$ b $\frac{1}{2}$ c $\frac{3}{4}$ d 1

41 Which of the following values of emf that is generated in the dynamo's coil during its rotation doesn't equal zero?

a The average emf $(emf)_{av}$ through a complete cycle.

b The average emf $(emf)_{av}$ through a half cycle from the parallel position with respect to the magnetic field.

c The instantaneous emf $(emf)_{inst}$ when the plane of the coil becomes parallel to the direction of the magnetic field.

d The instantaneous emf $(emf)_{inst}$ when the plane of the coil becomes perpendicular to the direction of the magnetic field.

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42 ✱ A dynamo's coil of 100 turns, the area of each is 200 cm^2, rotates in a magnetic flux such that it completes one revolution in 0.8 s where the average induced emf generated through $\frac{1}{4}$ cycle from the zero position equals 0.4 V. Then the magnetic flux density equals ..... .

(a) 0.01 T (b) 0.04 T (c) 0.16 T (d) 0.32 T

43 ✱ AC dynamo gives a maximum emf 100 V, so the average emf through a half cycle when the coil rotates from the perpendicular position equals ..... .

(a) 50 V (b) 70.7 V (c) 63.6 V (d) 100 V

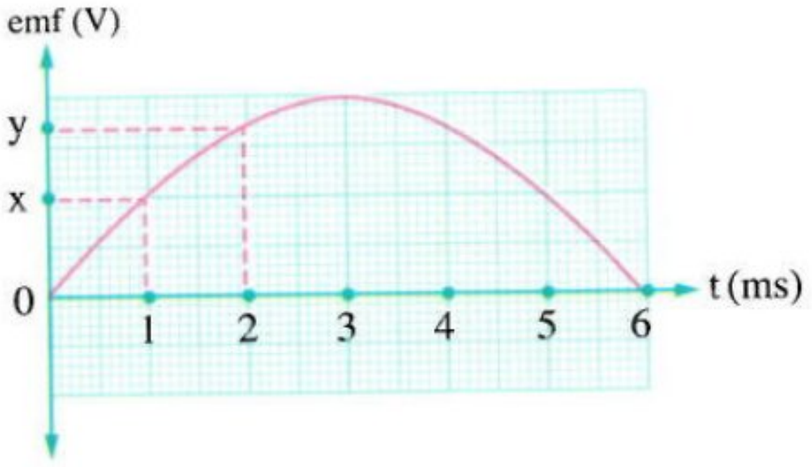
44 When the effective emf of the dynamo is 50 V, so the average emf within $\frac{1}{4}$ cycle from the parallel position to the field becomes almost ..... volts.

(a) 141.42 (b) 70.7 (c) 45 (d) 50

45 ✱ The opposite graph represents the change in the generated emf in an AC dynamo coil through half cycle, so the ratio $(\frac{x}{y})$ equals ..... .

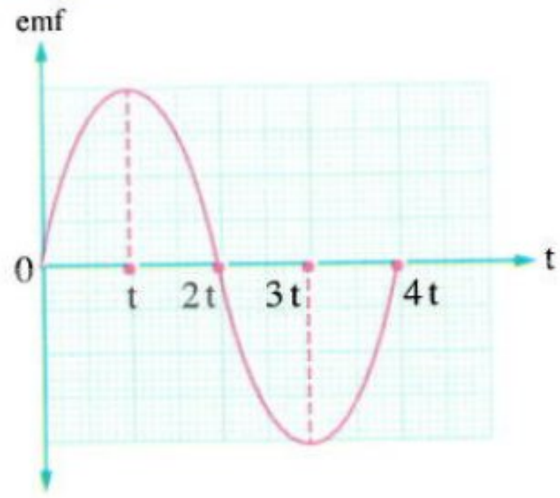
(a) $\frac{1}{2}$ (b) $\frac{\sqrt{2}}{2}$

(c) $\frac{1}{3}$ (d) $\frac{\sqrt{3}}{3}$



46 The opposite graph represents the relation between the induced (emf) in the dynamo coil and the time (t). In which interval of time an average emf equal to that generated in the interval from 0 to 2 t?

(a) The interval from t to 3 t
(b) The interval from t to 2 t
(c) The interval from 0 to 3 t
(d) The interval from 0 to 4 t



47 ✱ AC dynamo, its coil is formed of 100 turns, the area of each is 0.05 m^2, rotates in a magnetic field of flux density 0.1 T which generates a maximum induced emf of 157 V. Then: (Where: $\pi = 3.14$)

(i) The angular speed equals ..... .

(a) 157 rad/s (b) 314 rad/s (c) 471 rad/s (d) 511 rad/s

(ii) The frequency of the electric current generated in the coil equals ..... .

(a) 50 Hz (b) 75 Hz (c) 100 Hz (d) 120 Hz

(iii) The average induced emf through $\frac{1}{4}$ cycle from the position of maximum value equals ..... .

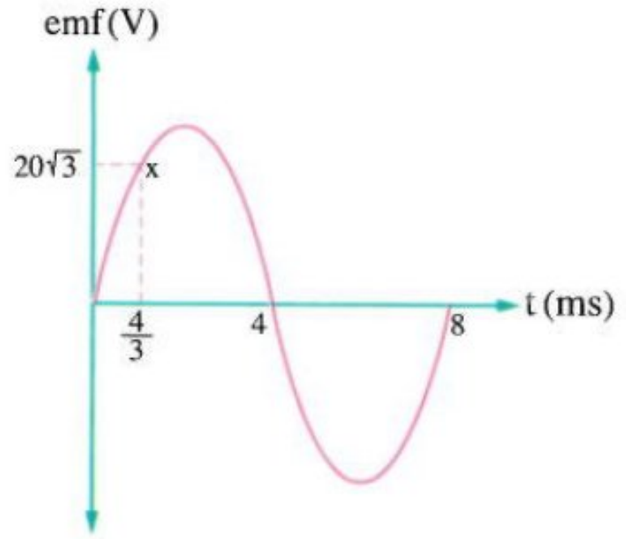
(a) 25 V (b) 50 V (c) 75 V (d) 100 V

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48 ✱ The opposite graph shows the relation between the instantaneous induced emf generated in a dynamo and the time (t), then:



(i) The maximum value of the induced emf in the dynamo's coil equals ..... .

(a) 80 V (b) 62.3 V
(c) 40 V (d) 34.6 V

(ii) The instantaneous value of emf for the AC current after $\frac{4}{3}$ ms from the instant at point x equals ..... .

(a) 29.96 V (b) $20\sqrt{3}$ V (c) 53.95 V (d) $40\sqrt{3}$ V

(iii) The effective value of the AC current intensity if the resistance of the dynamo's circuit is 10 Ω equals ..... .

(a) 2.446 A (b) 2.828 A (c) 4.404 A (d) 5.656 A

49 In the AC dynamo, the ratio between the average induced emf in the coil through $\frac{1}{4}$ cycle from the perpendicular position and the instantaneous emf after $\frac{1}{4}$ cycle from the perpendicular position $\left(\frac{(emf)_{av}}{(emf)_{inst}}\right)$ is ..... .

(a) $\frac{\pi}{1}$ (b) $\frac{\pi}{2}$ (c) $\frac{1}{\pi}$ (d) $\frac{2}{\pi}$

50 ✱ When an AC dynamo rotates $\frac{1}{6}$ cycle from the parallel position to the magnetic field, the average induced emf in the coil becomes emf_o, so when the dynamo rotates $\frac{1}{6}$ cycle from the perpendicular position to the magnetic field, the average induced emf in the coil becomes ..... .

(a) $\frac{emf_o}{3}$ (b) $3\ emf_o$ (c) $\sqrt{3}\ emf_o$ (d) $\frac{emf_o}{\sqrt{3}}$

51 An AC dynamo has a coil that rotates about its longitudinal axis so that it generates an induced electromotive force that varies according to the equation; emf = 240 sin (50 πt), so:

(i) The effective electromotive force is approximately equal to ..... .

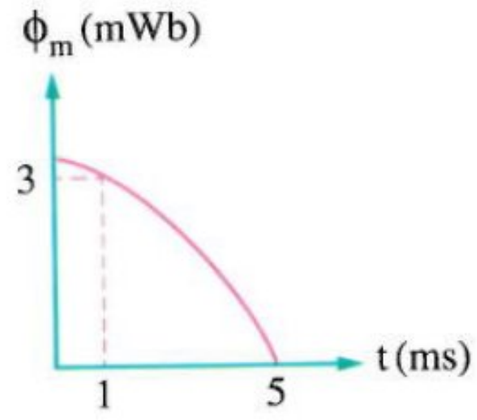
(a) $90\sqrt{2}$ V (b) $108\sqrt{2}$ V (c) $120\sqrt{2}$ V (d) $150\sqrt{2}$ V

(ii) The periodic time for the rotation of the coil equals .....

(a) 0.02 s (b) 0.03 s (c) 0.04 s (d) 0.06 s

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52 ✱ An AC dynamo coil of resistance 5 Ω, consists of 600 turns and connected to an external circuit of resistance 5 Ω, the opposite graph represents the relation between the magnetic flux (ϕ_m) which is passing through the coil and the time (t) through quarter of a cycle, so value of the current passing in the coil of the dynamo at any moment can be calculated from the relation; ..... (Where: $\pi = 3.14$)



(a) $I = 59.4 \sin (50 \pi t)$
(b) $I = 118.8 \sin (50 \pi t)$
(c) $I = 59.4 \sin (100 \pi t)$
(d) $I = 118.8 \sin (100 \pi t)$

53 ✱ AC dynamo coil of length 40 cm, width 30 cm and number of turns 300 turns, generates a current of frequency $\frac{50}{11}$ Hz where the effective value of the induced emf is $200\sqrt{2}$ V:

(i) So, the magnetic flux density affecting the coil equals ..... .

(a) $\frac{7}{18}$ T (b) 0.549 T (c) 0.274 T (d) $\frac{7}{36}$ T

(ii) If the speed of the dynamo's coil rotation is changed so that the linear speed of the longer side of the coil becomes 3 m/s, the maximum induced emf will be ..... .

(a) 140 V (b) 197.2 V (c) 280 V (d) 395.2 V

54 ✱ AC dynamo has coil consisting of 200 turns, the area of each is 2×10^{-2} m^2. It rotates in a magnetic field of flux density 0.1 T which generates emf of effective value 88.8 V. then: (Where: $\pi = 3.14$)

(i) The maximum value for the emf equals ..... .

(a) 88.8 V (b) 125.6 V (c) 177 V (d) 189.3 V

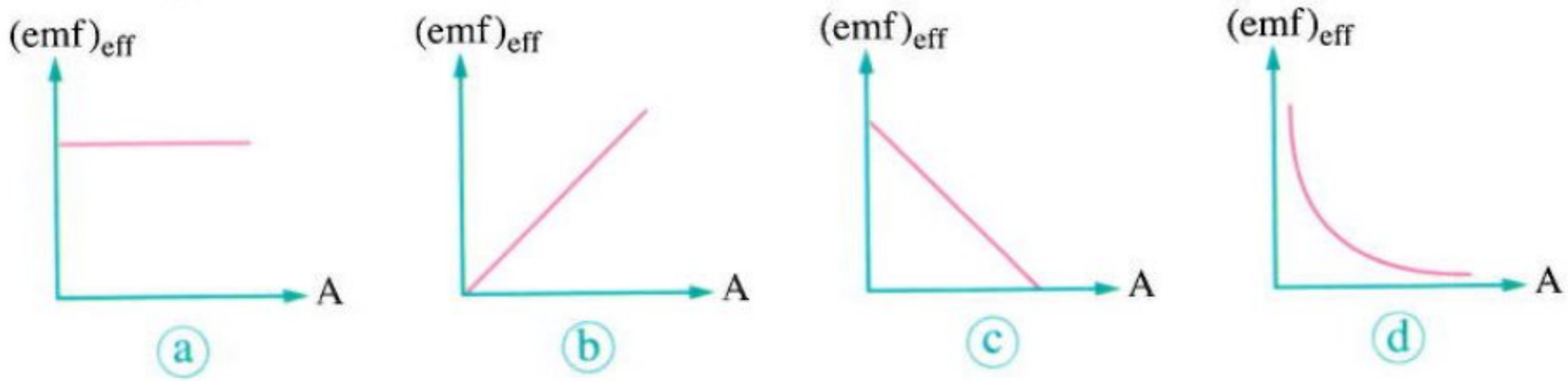
(ii) The current frequency equals ..... .

(a) 50 Hz (b) 110 Hz (c) 150 Hz (d) 200 Hz

55 ✱ The effective value of the emf which is generated from an AC dynamo is 100 V, then the average value of the emf during half cycle from the zero position is approximately equal to ..... .

(a) 141.42 V (b) 70.7 V (c) 90 V (d) 50 V

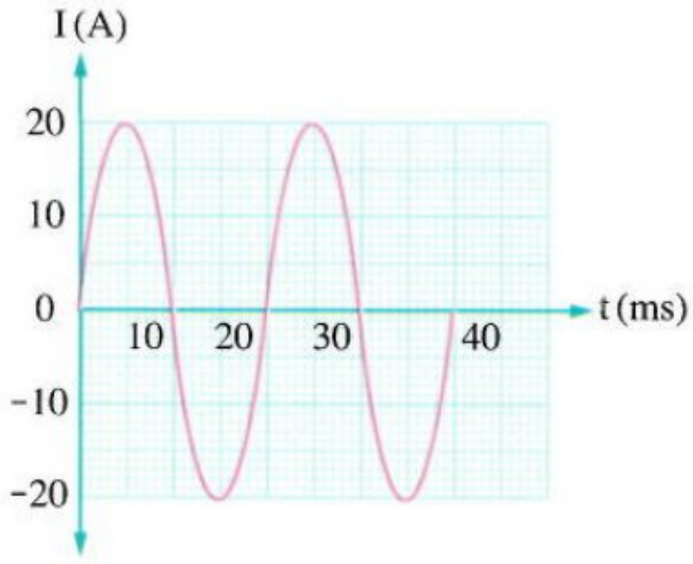
56 Which of the following graphs represents the relation between the value of the effective emf $(emf)_{eff}$ that is induced in the coil of a dynamo and the area of the coil (A)?



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57 The opposite figure represents the change in the electric current which is generated from an AC dynamo with time, then ..... .



	The angular speed	The effective value of the current
a	280.4 rad/s	20 A
b	280.4 rad/s	$10\sqrt{2}$ A
c	314.29 rad/s	20 A
d	314.29 rad/s	$10\sqrt{2}$ A

58 ✱ If the effective current intensity in an electric circuit (I_{eff}) is 2.828 A, then:

(i) The maximum value of the current (I_{max}) equals ..... .

a 2 A b 4 A c 6 A d 8 A

(ii) The instantaneous induced electric current intensity when the confined angle (θ) between the direction of the linear coil velocity and the direction of the magnetic flux density is 30° equals ..... .

a 2 A b 4 A c 6 A d 8 A

59 ✱ AC current has an effective value 3.535 A and frequency 50 Hz, then:

(i) The instantaneous value of the electric current intensity when the coil makes angle 60° with the magnetic flux equals ..... .

a 2.5 A b 5 A c 15 A d 25 A

(ii) The instantaneous electric current intensity after $\frac{1}{200}$ s from the zero position equals ..... .

a 5 A b 15 A c 25 A d 30 A

60 ✱ If the effective electric current intensity in AC circuit is 10 A, then the instantaneous current intensity:

(i) After $\frac{1}{4}$ cycle from the zero position equals ..... .

a $\frac{10}{\sqrt{2}}$ A b $10\sqrt{2}$ A c $\frac{\sqrt{2}}{10}$ A d 10 A

(ii) After $\frac{1}{8}$ cycle from the zero position equals ..... .

a $5\sqrt{2}$ A b 10 A c $5\sqrt{6}$ A d $10\sqrt{2}$ A

61 ✱ An AC source of maximum emf 200 V is connected to a resistance of 50 Ω, then:

(i) The maximum value of the current intensity equals ..... .

a $\frac{4}{\sqrt{2}}$ A b 3 A c 4 A d 5 A

(ii) The effective value of current intensity equals ..... .

a 1.727 A b 3.571 A c 2.828 A d 4.656 A

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62 The effective value of the AC current that is generated from a dynamo can be increased by each of the following **except** ..... .

(a) increasing the speed of rotation of the dynamo's coil

(b) increasing the number of turns of the dynamo's coil

(c) replacing the two metal rings by a cylindrical commutator that has two isolated metallic halves

(d) using stronger magnets

63 In the AC dynamo, if the time taken by the current to reach half maximum value $(\frac{I_o}{2})$ starting from the maximum value (I_o) is t_o, so:

(i) The current reaches $(\frac{I_o}{2})$ starting from zero for the first time through a time of ..... .

(a) $\frac{t_o}{2}$ (b) $\frac{t_o}{3}$ (c) $\frac{3}{2}\ t_o$ (d) $\frac{t_o}{4}$

(ii) The current reaches $(\frac{I_o}{2})$ starting from zero for the second time through a time of ..... .

(a) $\frac{3t_o}{2}$ (b) $\frac{4t_o}{3}$ (c) $\frac{5}{3}\ t_o$ (d) $\frac{5t_o}{2}$

(iii) The current reaches a value equals the effective value for the first time from zero through a time interval of ..... .

(a) $\frac{t_o}{2}$ (b) $\frac{t_o}{3}$ (c) $\frac{2}{3}\ t_o$ (d) $\frac{3t_o}{4}$

(iv) The periodic time of the current equals ..... .

(a) $4t_o$ (b) $6t_o$ (c) $8t_o$ (d) $12t_o$

64 ✱ If the time interval required for an alternating current to rise from zero to half its maximum value is t, then the time required to reach from zero to the maximum value is ..... .

(a) t (b) 2 t (c) 3 t (d) 4 t

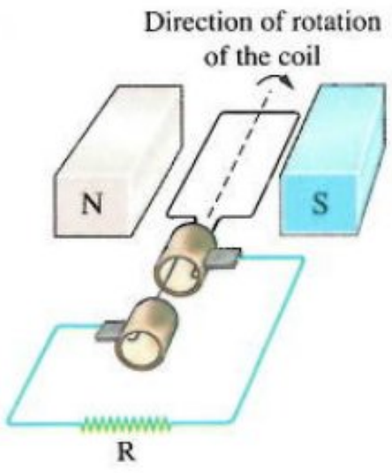
65 An AC dynamo rotates with a frequency of 25 Hz where the maximum value of the current passing in its external circuit equals I_o, so the current reaches $\frac{\sqrt{3}}{2}\ I_o$ from zero for the first time through a time interval of ..... .

(a) $\frac{1}{75}$ s (b) $\frac{1}{120}$ s (c) $\frac{1}{150}$ s (d) $\frac{1}{300}$ s

66 Starting from the position that is shown in the opposite figure for an AC dynamo, through one and a half cycle of the coil, the current I that passes through the resistance R changes its direction ..... .

(a) twice (b) three times

(c) four times (d) five times



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67 ✳ AC electric generator rotates with a rate of 20 rotation every 0.4 s and generates a current of maximum value 5 A. Then: (Where: $\pi = 3.14$)

(i) The number of times the value of current reaches 5 A during 1 s starting from zero position is ..... .

(a) 50 times (b) 75 times (c) 100 times (d) 200 times

(ii) The number of times the value of current reaches zero during 1 s starting from zero position is ..... .

(a) 100 times (b) 101 times (c) 102 times (d) 105 times

68 An AC dynamo has a coil that rotates with a frequency of 100 Hz. If the rotation of the coil has started to be observed from the position at which the coil was parallel to the magnetic field direction, the number of times at which the current reaches a value equal to its effective value during 1 s is ..... .

(a) 200 times (b) 201 times (c) 400 times (d) 401 times

69 ✳ An AC dynamo coil started rotation with uniform speed from the parallel position to the field, if the direction of the current passing in the coil is reversed n times within time interval t, so the frequency of rotation of the dynamo equals ..... .

(a) $\frac{n}{t}$ (b) $\frac{2n}{t}$ (c) $\frac{n}{2t}$ (d) $\frac{n+1}{2t}$

70 ✳ A dynamo coil consists of 100 turns, each of resistance 0.01 Ω, when it rotates with a frequency of 50 Hz while connecting its terminals together, it consumes electric energy of 2 J within one cycle, then the maximum induced emf equals ..... .
(Where: $\pi = 3.14$)

(a) 7.07 V (b) 14.14 V (c) 21.21 V (d) 28.28 V

71 ✳ In an AC dynamo of negligible resistance, the instantaneous induced emf is given by the relation; emf = 200 sin 18000 t. Then: (Where: $\pi = 3.14$)

(i) The current frequency equals ..... .

(a) 200 Hz (b) 150 Hz (c) 100 Hz (d) 50 Hz

(ii) The magnitude of emf after 5 ms elapsed from the position at which the plane of the coil was perpendicular to the magnetic field equals ..... .

(a) 50 V (b) 100 V (c) 150 V (d) 200 V

(iii) The consumed electric energy in a resistance of 20 Ω that is connected to the external circuit of the dynamo during one cycle of the alternating current equals ..... .

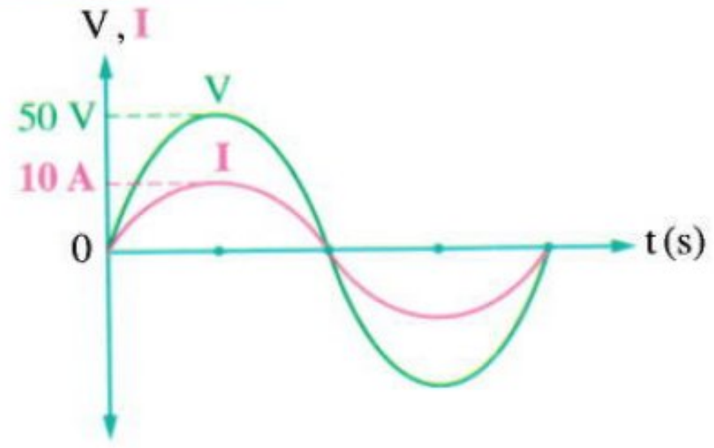
(a) 20 J (b) 40 J (c) 60 J (d) 80 J

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72 ✱ AC dynamo was connected to a resistor of 8 Ω that produced thermal energy of 200 J within a time interval of 1 s, then the maximum value for both of the electric current intensity through the resistor and the potential difference across its terminals respectively are ....., ...

(a) 5.072 A, 45.87 V
(b) 5.072 A, 56.58 V
(c) 7.072 A, 50.72 V
(d) 7.072 A, 56.58 V

73 The opposite graph represents the variation of the potential difference and the current in the circuit of an electric generator on the vertical axis versus the time (t) on the horizontal axis, so the generated electric power equals ..... .



(a) 50 W
(b) 125 W
(c) 250 W
(d) 500 W

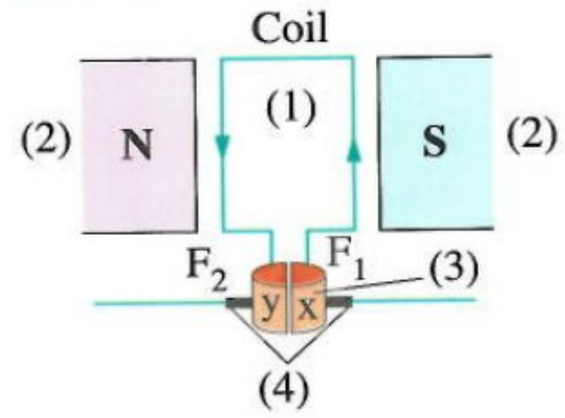
74 The ratio between the number of coils to the number of segments of the hollow metallic cylinder in the unidirectional dynamo equals ..... .

(a) $\frac{1}{2}$
(b) $\frac{1}{1}$
(c) $\frac{2}{1}$
(d) $\frac{4}{1}$

75 If the frequency of the generated current from an AC dynamo is 50 Hz, so the frequency of the current of that dynamo after being rectified into a unidirectional current will equal ..... .

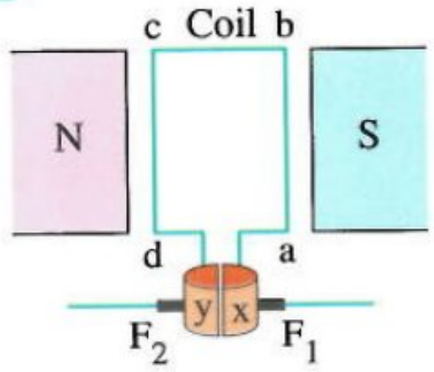
(a) 25 Hz
(b) 50 Hz
(c) 100 Hz
(d) 200 Hz

76 The opposite figure represents the structure of a unidirectional electric current dynamo, then the responsible component for the rectification of the alternating current is ..... .



(a) component (1)
(b) component (2)
(c) component (3)
(d) component (4)

77 The opposite diagram represents the structure of a dynamo. If side ab moves at this moment out of the page and the coil of the dynamo has rotated a complete cycle, so the brush ..... .

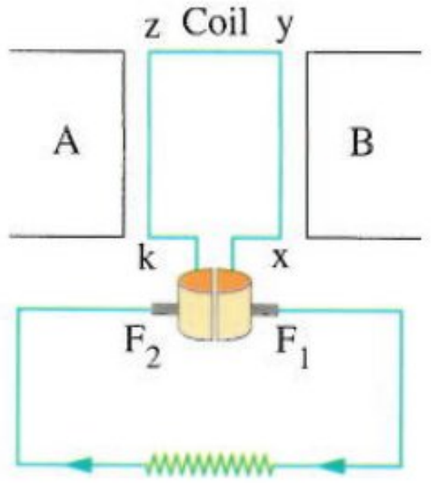


(a) F_1 always works as a positive terminal during the two halves of the cycle
(b) F_2 always works as a positive terminal during the two halves of the cycle
(c) F_1 works as a positive terminal during one of the halves of the cycle
(d) F_2 works as a positive terminal during one of the halves of the cycle

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78 The opposite figure shows the structure of a dynamo of a unidirectional electric current and the direction of that current, so which of the following choices represents the types of magnet poles A, B and direction of motion of the side xy of the coil at this moment?

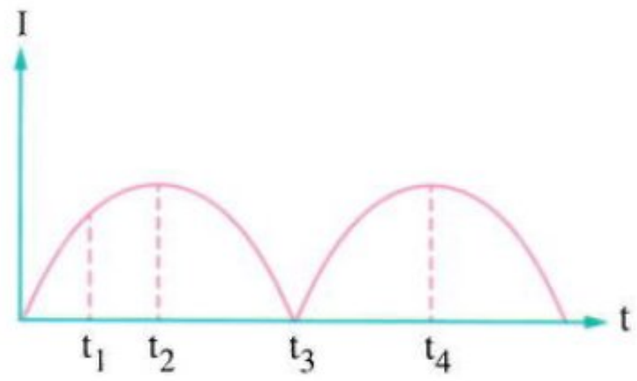


	A	B	The direction of motion of xy side
(a)	N	S	out of the page
(b)	N	S	into the page
(c)	S	N	into the page
(d)	S	N	towards the pole B

79 The unidirectional current dynamo in which one coil is used is designed such that the angle between the plane of that coil and the plane of the insulating material in the splitted metal cylinder is ..... .

(a) 90°
(b) 60°
(c) 0
(d) changing with the rotation of the coil

80 The opposite graph represents the current produced from the dynamo, so:



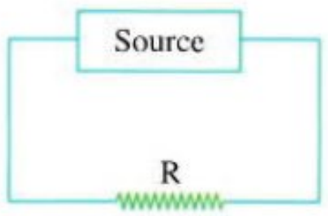
(i) The splitted cylinder is divided into number of parts equals ..... .

(a) two parts
(b) three parts
(c) four parts
(d) eight parts

(ii) The instant at which the graphite brushes touch the insulating material between the parts of the splitted cylinder is ..... .

(a) t_1 (b) t_2
(c) t_3 (d) t_4

81 The following graphs represent the probabilities of the relation between the current (I) passing in the resistor R and the time (t) in the opposite electric circuit.



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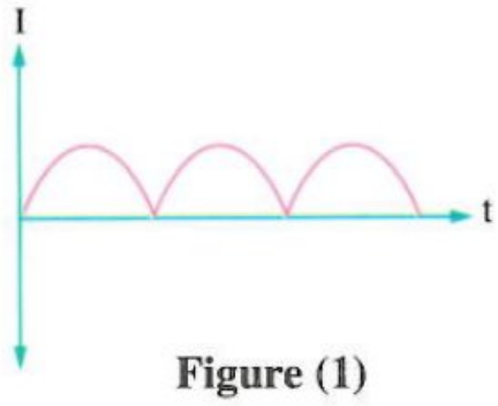


Figure (1)

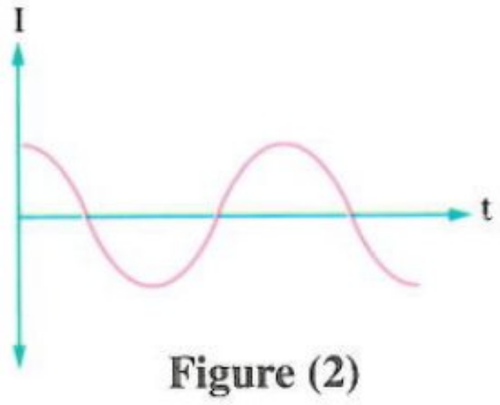


Figure (2)

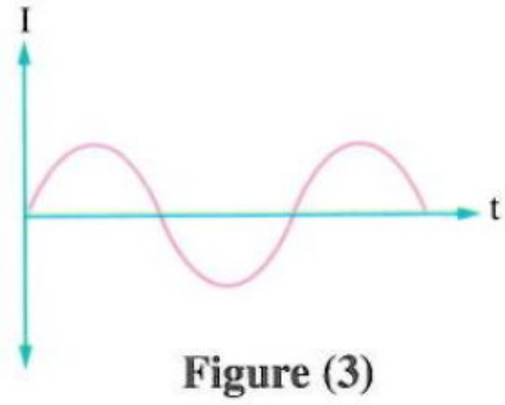


Figure (3)



Figure (4)

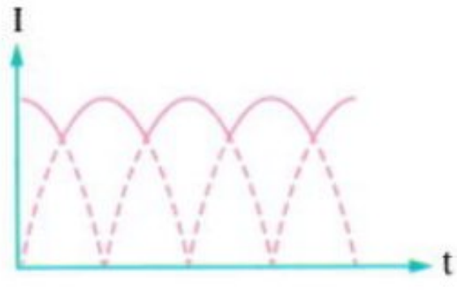


Figure (5)

Which of these graphs is obtained if the source is:

(i) AC dynamo started rotation from the perpendicular position to the magnetic flux?

(a) Figure 1 (b) Figure 2 (c) Figure 3 (d) Figure 5

(ii) AC dynamo started rotation from the parallel position to the magnetic flux?

(a) Figure 1 (b) Figure 2 (c) Figure 3 (d) Figure 4

(iii) Unidirectional current dynamo contains one coil?

(a) Figure 1 (b) Figure 2 (c) Figure 4 (d) Figure 5

(iv) Unidirectional current dynamo contains two perpendicular coils?

(a) Figure 1 (b) Figure 3 (c) Figure 4 (d) Figure 5

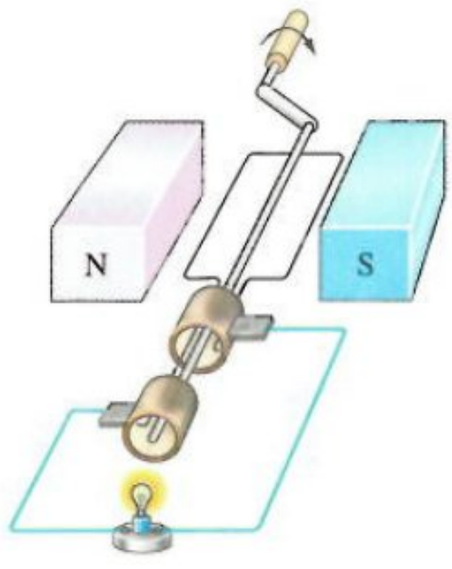
Second Essay questions

1 In the opposite figure:

(a) How can you explain the lighting up of the lightbulb when rotating the handle?

(b) Mention two physical quantities for the electric current through the bulb filament that increase with increasing the speed of rotating handle.

(c) What happens to the brightness of the bulb when rotating the dynamo coil in the opposite direction with the same angular speed? **And why**?



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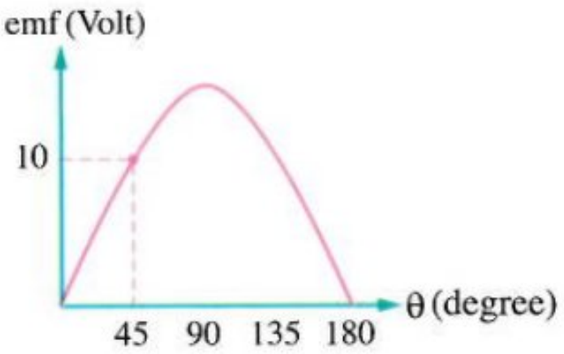
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2 An AC dynamo coil of 420 turns and area 3×10^{-3} m^2 rotates in a magnetic field of flux density 0.5 T. If the coil takes a time interval of $\frac{1}{200}$ s to rotate from the position in which its plane is perpendicular to the flux lines till reaching the position in which the maximum induced emf gets generated across it for the first time, **calculate:**

(a) The maximum induced emf.

(b) The time taken by the coil so that the induced emf in the coil changes from zero to half its maximum value.

3 The opposite graph shows the relation between the induced electromotive force (emf) in a dynamo coil and the angle (θ) between the normal to the coil plane and the direction of the magnetic flux, **so calculate** the maximum value of the induced electromotive force.



4 An AC dynamo has a coil of 100 turns, in which the induced emf rises from zero to a value equal to its effective value for the first time after 0.01 s, so if the maximum magnetic flux linked with the coil is 0.02 Wb, **calculate:**

(a) The maximum induced emf $(emf)_{max}$

(b) The effective emf $(emf)_{eff}$

5 If the instantaneous induced emf in an AC dynamo coil is given from the relation: $emf = 100\,\pi BA \sin(100\,\pi t)$, **calculate:**

(a) The dynamo coil rotation frequency.

(b) The dynamo coil number of turns.

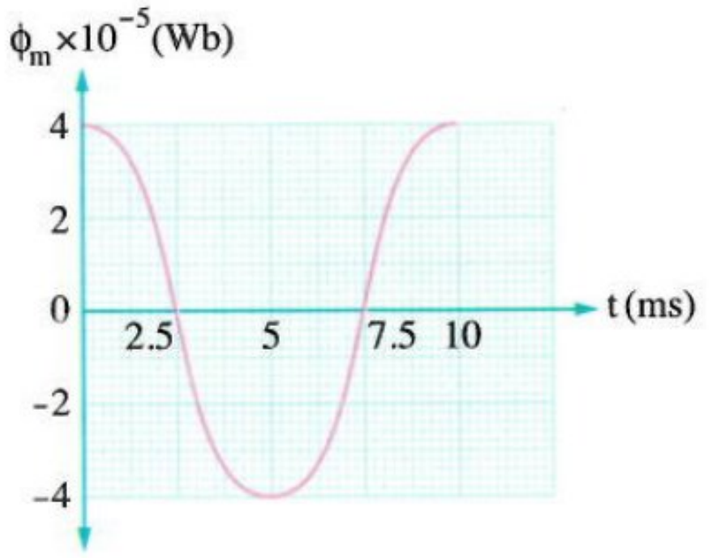
6 **Give reason:** Even though the average value of an AC current in a conductor of resistance R through a complete cycle equals zero, the consumed electric energy does not equal zero.

7 **Prove that:** The average of the induced emf in the coil of an AC dynamo through its rotation with an angle of 60° from the zero position is given by the relation; $(emf)_{av} = \frac{3}{2\pi}(emf)_{max}$

8 **Prove that:** The average induced emf in the dynamo coil through $\frac{1}{4}$ cycle = The average induced emf in the dynamo coil through $\frac{1}{2}$ cycle starting from zero position.

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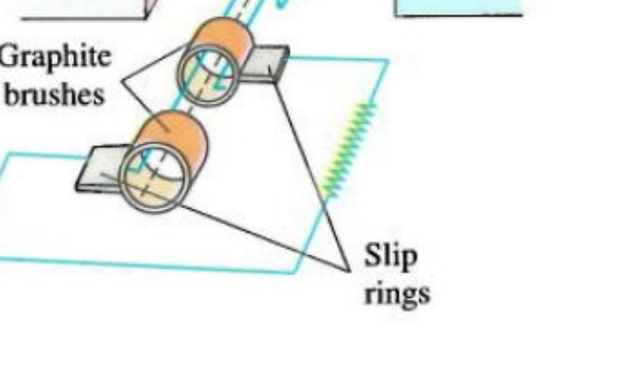
9 The opposite graph represents the variation of the magnetic flux (ϕ_m) through a coil of negligible resistance that consists of 150 turns while rotating with a constant angular speed (ω) in a magnetic field versus time (t), where its terminals are connected to a closed circuit of resistance 5 Ω, **so calculate** the average intensity of the induced current through the coil during the interval from 2.5 ms to 5 ms.



10 The opposite graph represents the change in the induced emf in the coil of an electric generator versus time:

(a) **Which** of the points (A, B, C or D) shown on the graph represents the induced emf in the coil when it passes by the perpendicular position to the field?

(b) **Calculate** the elapsed time for the induced emf in the coil to reach from 45 V which is denoted by point A to 22.5 V for the first time.



11 The opposite figure illustrates the structure of an AC dynamo rotating with a frequency of 10 Hz, so starting from the position in which the dynamo coil is perpendicular to the magnetic field, **calculate the required time taken by the coil to reach the following states for the first time:**

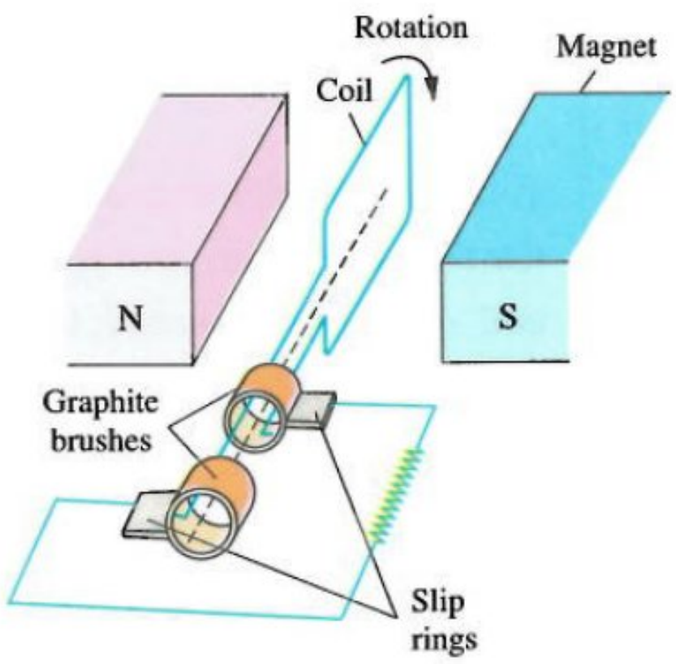
(a) The magnetic flux passing in the coil vanishes.

(b) The induced emf in the coil becomes maximum.

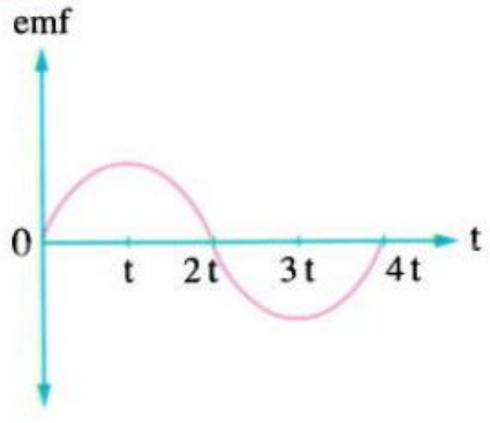
(c) The alternating current intensity generated in the coil becomes zero.

(d) The average of the induced emf in the coil = zero

(e) The instantaneous emf induced in the coil becomes equal to the effective emf which is produced from it.



12 The opposite graph represents the relation; (emf - t) for a coil of an AC dynamo. **Show that** the average induced emf in the coil through the time interval from 0 to t is greater than the average induced emf through the time interval from 0 to 3 t.



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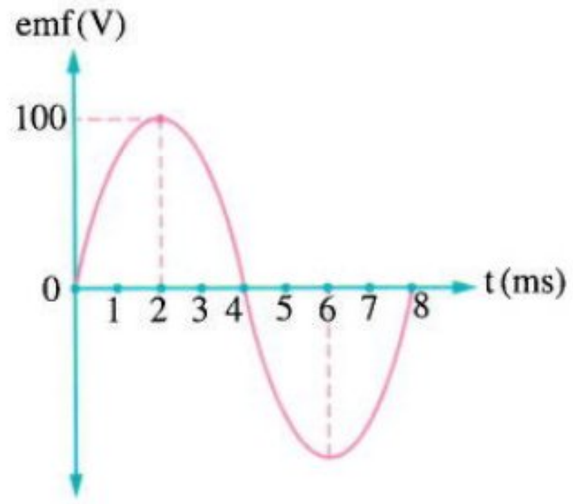
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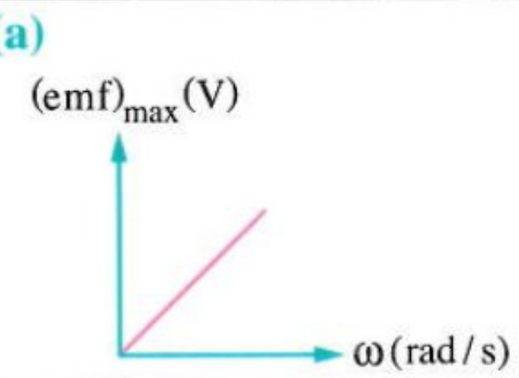
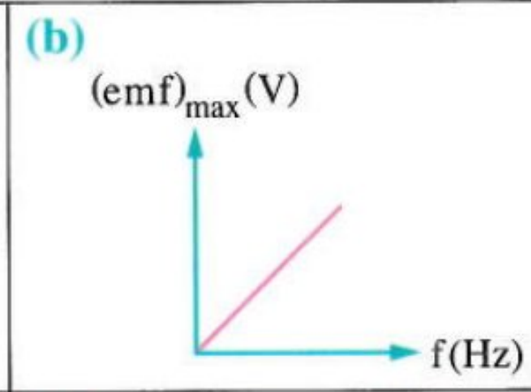
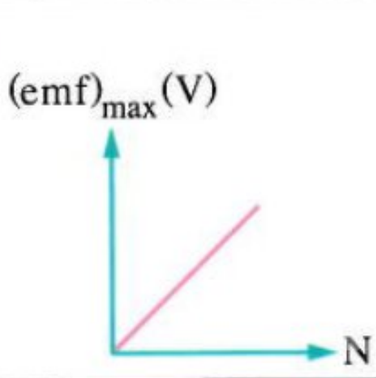
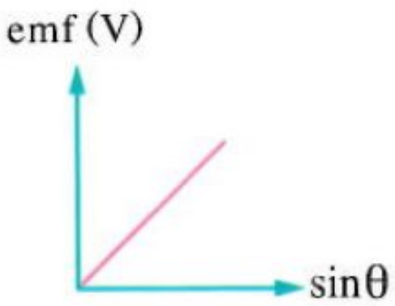
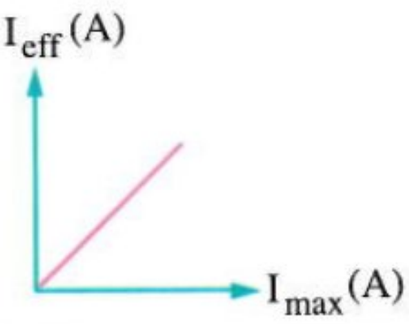
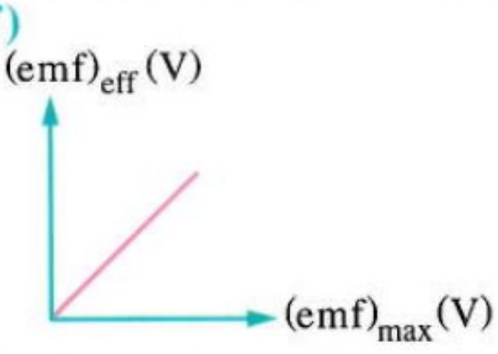
13 The opposite graph represents the curve of the induced emf in an AC dynamo coil, **find the average of emf through the following time intervals:**

(a) From t = 2 ms to t = 6 ms

(b) From t = 3 ms to t = 4 ms



14 **Write the mathematical relation and the equivalent to the slope for each of the following graphs:**

(a) 	**(b)** 	**(c)** 
(d) 	**(e)** 	**(f)** 

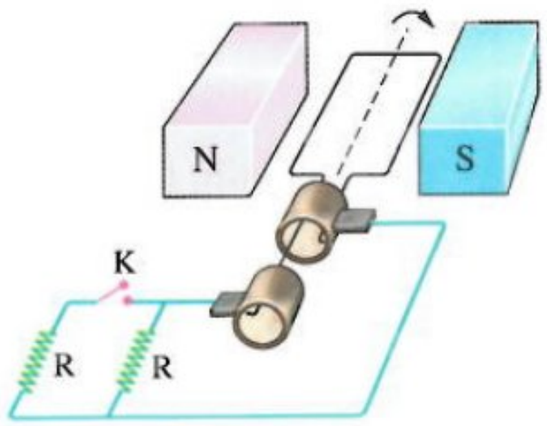
Where: (ω) is the angular velocity, (θ) is the angle between the normal to the coil plane and the field direction, $(emf)_{max}$ is the maximum induced emf, (N) is the number of turns of the coil, (f) is the frequency, $(I)_{eff}$ is the effective value of current intensity, $(I)_{max}$ is the maximum value of the current and $(emf)_{eff}$ is the effective value of emf.

15 **Explain what happens when** increasing the number of turns of a dynamo coil to the double with increasing the number of revolution of the coil per second also to the double, concerning the maximum and the effective values of the induced emf in the coil.

16 The opposite figure represents an AC dynamo rotating with a uniform rate, so if switch K is closed while keeping the dynamo coil rotating with the same frequency, **explain what happens to:**

(a) the maximum value of the induced emf in the coil.

(b) the maximum value of the current through the coil.

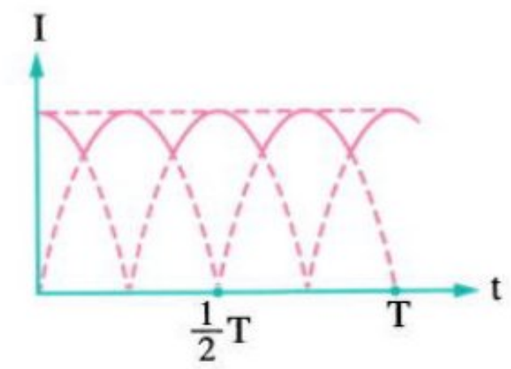


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17 **Explain why** when connecting an electric bulb to the terminals of a 12 V battery of negligible internal resistance, the bulb glows brighter than when it gets connected to the terminals of a dynamo of maximum voltage 12 V and negligible internal resistance.

18 **What are the types of** each of the current induced in the dynamo coil and the current passing in its external circuit when the dynamo coil terminals are connected to a split metal cylindrical commutator of two isolated halves?

19 The opposite graph represents the relation between the current value (I) in the external dynamo circuit and time (t):



(a) **How many** coils does this dynamo have?

(b) **How can** this dynamo be modified to get a current of approximately constant intensity?

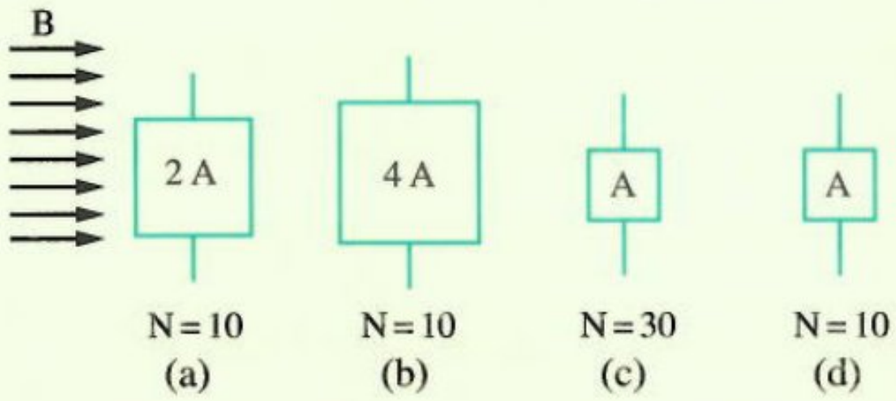
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Questions of the Previous Years Exams on Lesson 3

1



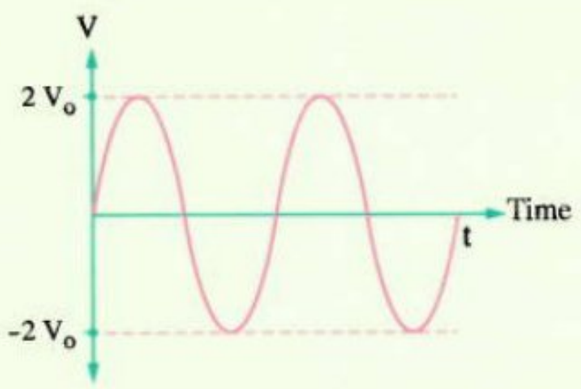
The figure represents four different coils where their cross-sectional areas and their number of turns are indicated on the figure. All coils rotate with the same angular velocity about an axis that is normal to the direction of a magnetic field (B), so the ascending arrangement of the coils according to the maximum value of the induced emf in each of them is ..... *(Exp. May 21)*

(a) b ⟶ c ⟶ a ⟶ d
(b) d ⟶ a ⟶ c ⟶ b
(c) c ⟶ b ⟶ a ⟶ d
(d) a ⟶ d ⟶ b ⟶ c

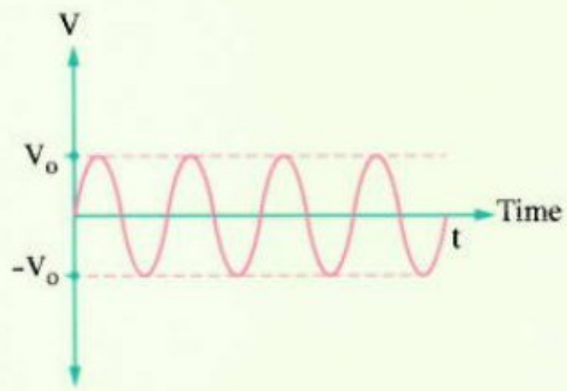
2 An AC dynamo of 100 turns and a cross-sectional area of 250 cm^2 is rotating in a magnetic field of flux density 200 mT, starting from the perpendicular position to the field, the generated emf in the dynamo reaches its maximum value 100 times in one second, so the value of the effective induced emf = ... *(Exp. May 21)*

(a) 157.1 V (b) 111.1 V (c) 222.2 V (d) 314.3 V

3 Each graph represents the number of voltage cycles generated by different dynamos (x) and (y) during the same time interval (t).



The produced voltage from dynamo's coil (x)



The produced voltage from dynamo's coil (y)

If the two coils of the dynamos (x) and (y) have the same cross-sectional area and rotate in a magnetic field of the same strength, the ratio of $\frac{\text{Number of turns in coil (y)}}{\text{Number of turns in coil (x)}}$ = ... *(Exp. May 21)*

(a) $\frac{1}{6}$ (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

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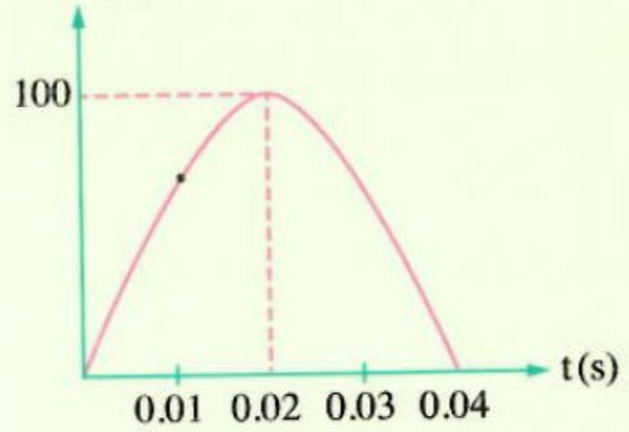
4 A coil of AC generator consists of 12 turns, each of face area A = 0.08 m^2. The total resistance of the coil's wire is 22 Ω and the coil rotates in 0.6 T magnetic field to produce an AC of frequency 50 Hz, then the maximum induced current from the dynamo when its output terminals are connected to a negligible resistance conductor equals ..... . *(Exp. May 21)*

(a) 8.23 A (b) 11.8 A (c) 18.5 A (d) 23.4 A

5 A simple electric generator lights an electric bulb of power 60 W and resistance 30 Ω, hence the maximum electric current intensity passing in the bulb equals ..... .

(a) 2 A (b) $\sqrt{2}$ A (c) 1 A (d) 0.5 A *(Exp. June 21)*

6 The graph represents the relation between the induced electromotive force (emf) in a dynamo's coil and time (t) through half cycle, so the average electromotive force generated in the dynamo's coil during the interval of time from t = 0 to t = $\frac{1}{75}$ s equals ..... volt. (Take: π = 3.14) *(Exp. June 21)*

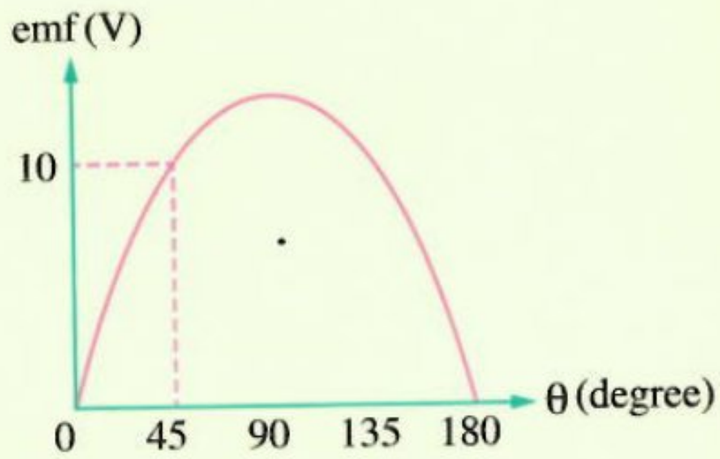


(a) 47.77 (b) 63.69 (c) 21.23 (d) 86.603

7 A simple electric generator has a coil of area 0.02 m^2 that rotates in a magnetic field of flux density 0.1 T with a rate of 50 rotations in a second. If the coil has 100 turns of wire, the average electromotive force that is induced during a half cycle starting from the perpendicular position to the field equals ..... . *(1st session 21)*

(a) 20 V (b) 10 V (c) 40 V (d) 30 V

8 The opposite graph represents the change of the induced electromotive force (emf) in the dynamo's coil with respect to the angle between the normal to the coil's plane and the direction of the magnetic flux (θ), so the magnitude of the average induced emf in the dynamo's coil during $\frac{1}{3}$ cycle from the starting of the coil's rotation equals ..... . *(1st session 21)*



(a) 3.002 V (b) 6.369 V (c) 9.006 V (d) 10.1 V

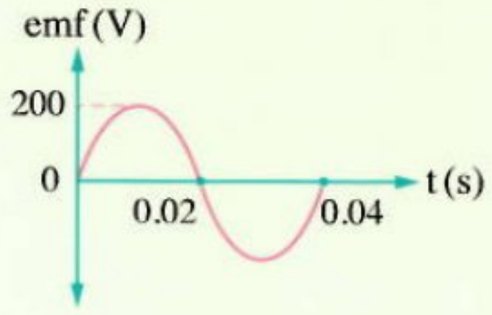
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9 In a simple electric generator, its coil starts rotation from the vertical position, if the instantaneous electromotive force induced in the coil reaches half of the maximum value of the induced emf for the second time after $\frac{1}{60}$ s from the beginning of the coil rotation, so the frequency of the produced current equals ..... . *(2nd session 21)*

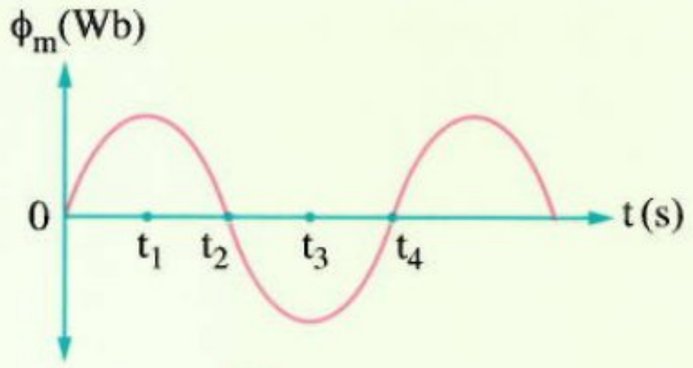
(a) 5 Hz (b) 50 Hz (c) 25 Hz (d) 15 Hz

10 The graph represents the relation between the induced electromotive force (emf) in the dynamo and time (t). From the figure, the average induced electromotive force of the dynamo coil through interval of time from t = 0 to t = $\frac{1}{30}$ s equals ..... . (Where: $\pi = 3.14$) *(2nd session 21)*



(a) 127.4 V (b) 42.5 V (c) 173.2 V (d) 19.1 V

11 The opposite graph represents the relation between the change of magnetic flux through a rectangular coil with time, so the instantaneous induced electromotive force equals zero at the points ..... . *(2nd session 21)*

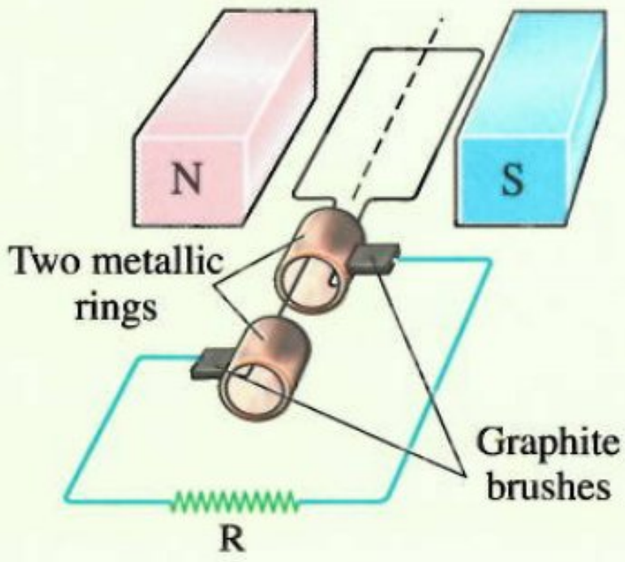


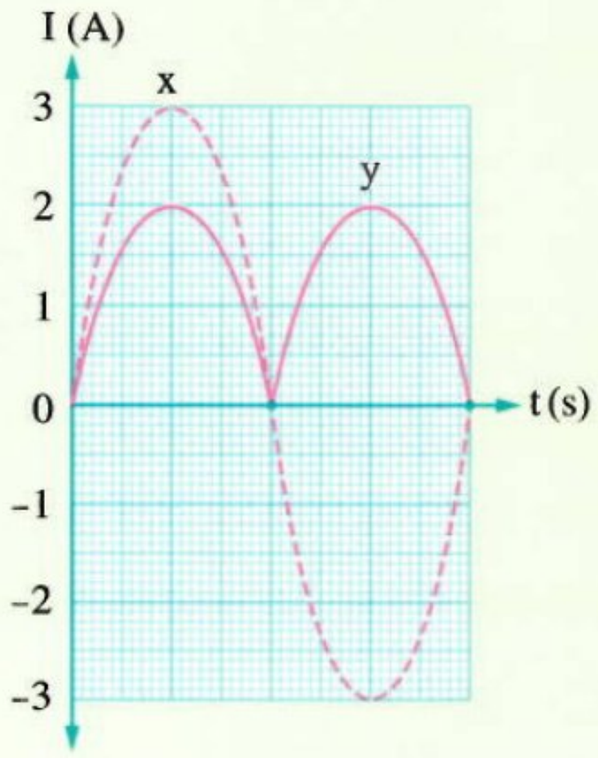
(a) t_1, t_3 (b) t_2, t_4 (c) t_1, t_2 (d) t_1, t_4

12 AC dynamo whose coil consists of 200 turns and has area of 0.01 m^2 rotates inside a magnetic field of flux density 0.3 T. If the maximum value of the produced e.m.f is 376.99 V, the angular velocity of the coil equals ..... rad/s. ($\pi = 3.14$)

(a) 100 π (b) 50 π (c) 150 π (d) 200 π *(1st session 22)*

13 A student drew the sine-curve of the produced current from the shown dynamo (its coil has resistance of 10 ohms) by two different curves (x) and (y).





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From the curve that correctly represents the produced current, the average induced emf that was produced within half cycle starting the rotation from the perpendicular position equals ..... . ($\pi = 3.14$) *(1st session 22)*

(a) 12.74 V (b) 19.11 V (c) 4.78 V (d) 3.18 V

14 A coil is placed inside a uniform magnetic field where its plane is perpendicular to the flux lines direction, then the ratio between:

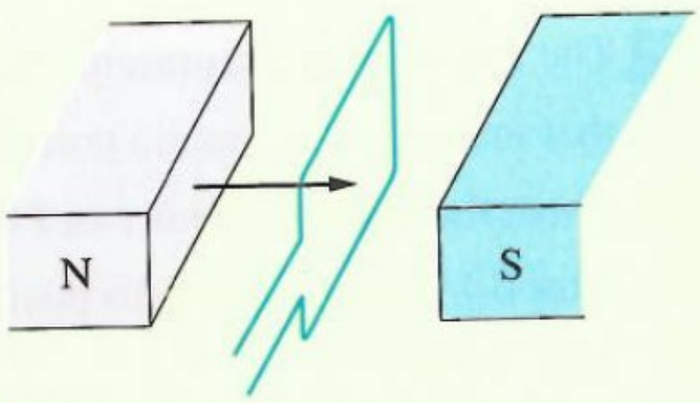
$$\frac{\text{The average induced e.m.f in the coil when it is rotated } \frac{1}{4} \text{ cycle during time (t)}}{\text{The average induced e.m.f in the coil when it is rotated } \frac{1}{2} \text{ cycle during the same time (t)}} = \text{.....}$$

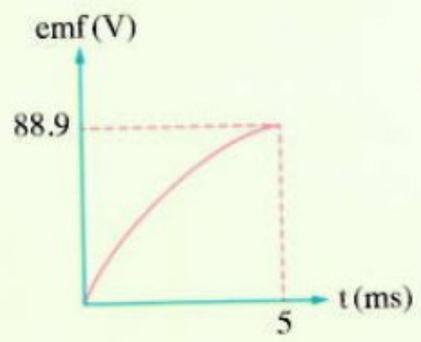
(a) 0.5 (b) 1 (c) 0.25 (d) 0.75 *(2nd session 22)*

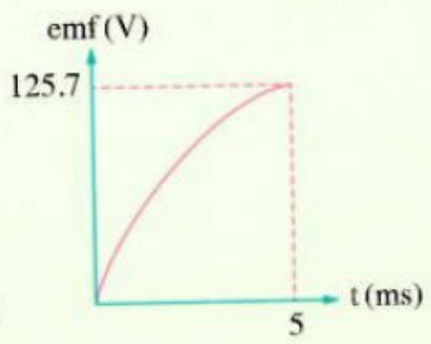
15 A dynamo coil starts rotation from the perpendicular position at a frequency of 50 Hz, it produces a maximum electromotive force of 100 V. So, the time needed for the instantaneous induced electromotive force to reach 50 V for the second time from the beginning equals ..... . *(2nd session 22)*

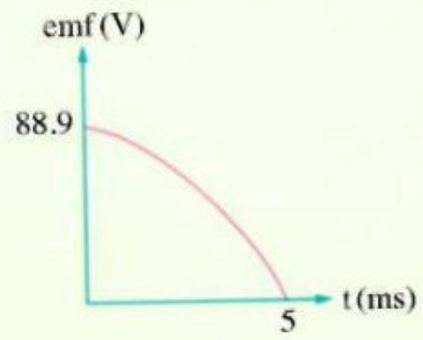
(a) $\frac{1}{600}$ s (b) $\frac{1}{400}$ s (c) $\frac{1}{120}$ s (d) $\frac{1}{200}$ s

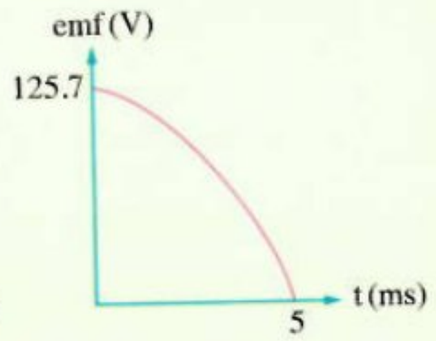
16 The figure shows a dynamo coil of 200 turns and area 0.1 m^2 that rotates starting from the perpendicular position with frequency 50 Hz between two poles of a magnet of flux density 20 mT. Which of the following graphs correctly represents the produced instantaneous emf in the dynamo coil from 0 ms to 5 ms? *(2nd session 22)*



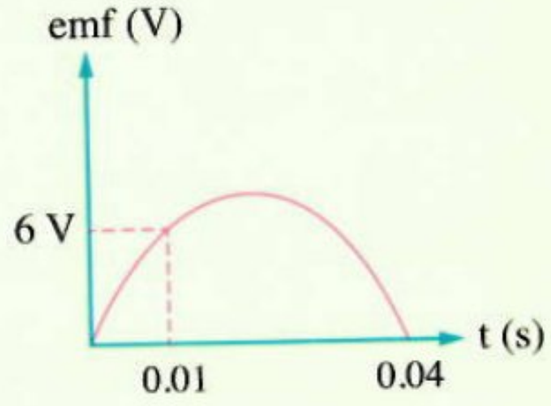
(a)


(b)


(c)


(d)


17 The graph illustrates the relation between the induced electromotive force (emf) of dynamo coil and the time (t), the effective value of induced electromotive force is ..... . *(Exp. 23)*



(a) 6 V (b) $6\sqrt{2}$ V

(c) 12 V (d) $12\sqrt{2}$ V

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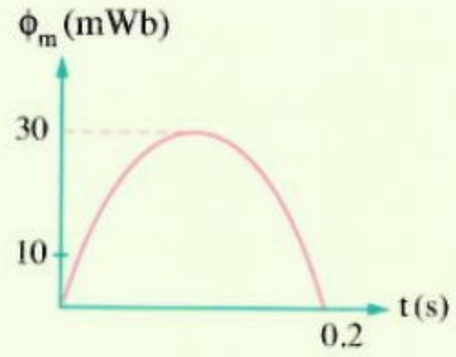
18 An alternating current dynamo, the area of the face of its coil is 0.02 m^2 and the number of turns of that coil is 300 turns, rotates with a rate of 1400 cycle/minute in a field of magnetic flux density 0.01 T, so the instantaneous induced electromotive force in the coil when the coil plane makes angle 60° with the direction of magnetic field lines is ..... .

(a) 8.8 V (b) 4.4 V (c) 7.62 V (d) 2.2 V *(Exp. 23)*

19 AC dynamo, its coil consists of 200 turns each of face area 0.02 m^2, rotates in a magnetic field of flux density 0.02 T by a rate of 6000 rotations per minute, then the effective value of the induced e.m.f. equals ..... . (Knowing that: $\pi = 3.14$) *(1st session 23)*

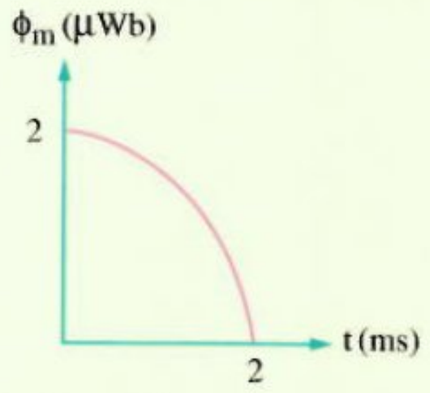
(a) 35.53 V (b) 25.12 V (c) 17.76 V (d) 12.56 V

20 The graph represents the relation between the change in the magnetic flux (ϕ_m) that is intercepting a coil versus time (t). If you know that the coil consists of 200 turns, then the average induced e.m.f in the coil within the time interval of 0.2 s represented in the graph equals ..... . *(1st session 23)*



(a) 0 V (b) 60 V (c) 30 V (d) 45 V

21 The shown graph represents the change of magnetic flux (ϕ_m) that intercepts a dynamo coil of 200 turns and the time (t). Then, the instantaneous e.m.f that is produced in the coil after 0.1 ms from the zero position equals ..... . ($\pi = 3.14$) *(1st session 23)*

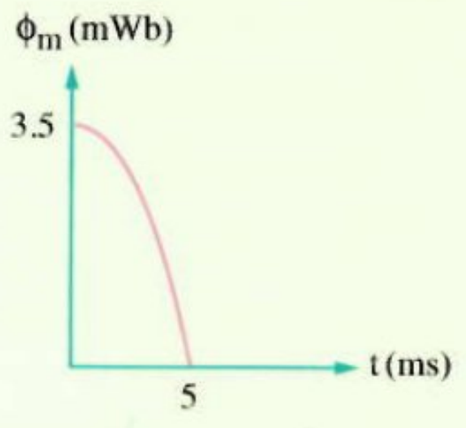


(a) 0.0025 V (b) 0.25 V (c) 0.025 V (d) 0.00025 V

22 An alternating current dynamo produces a current of frequency 50 Hz, so the time for the current to reach a value equal to its effective value for the first time from the perpendicular position = ... *(2nd session 23)*

(a) 0.5 ms (b) 1.5 ms (c) 2.5 ms (d) 0.25 ms

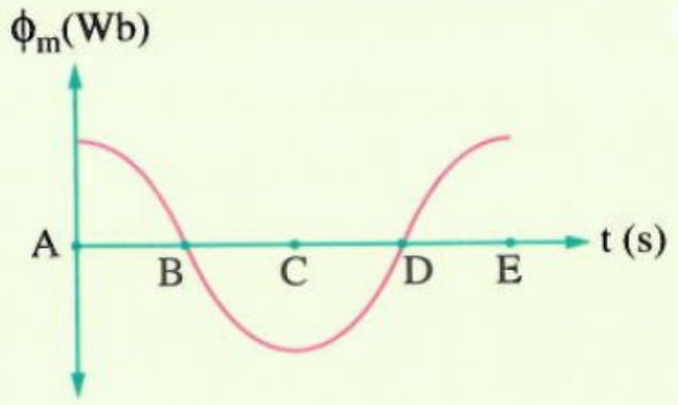
23 The graph represents the change in the magnetic flux (ϕ_m) that intercepts a dynamo coil of 200 turns versus time (t). So, the average induced e.m.f. in the dynamo coil during the shown $\frac{1}{4}$ cycle equals ..... . *(2nd session 23)*



(a) 155.56 V (b) 220 V
(c) 140 V (d) 110 V

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24 The graph represents the change in the magnetic flux (ϕ_m) which penetrates the dynamo coil during its rotation with time, so the instantaneous induced emf in the coil equals ..... . *(1st session 24)*



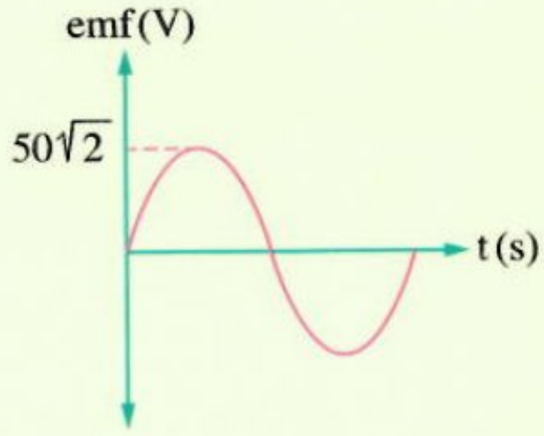
(a) zero at the moments B, D

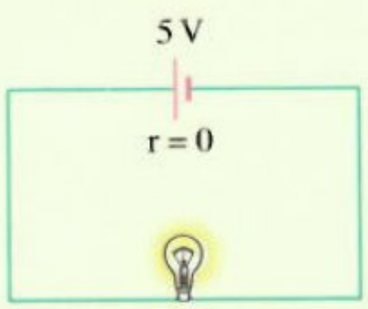
(b) maximum value at the moments D, C

(c) zero at the moments A, C

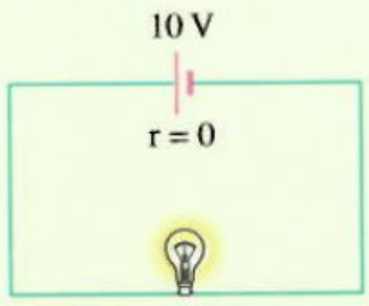
(d) maximum value at the moments B, C

25 The opposite graph represents the relation between the induced emf in an AC dynamo that its circuit resistance is 500 Ω and time (t). So, the effective value of the passing current in the dynamo was (I). In which of the following circuits a current of intensity (I) passes?

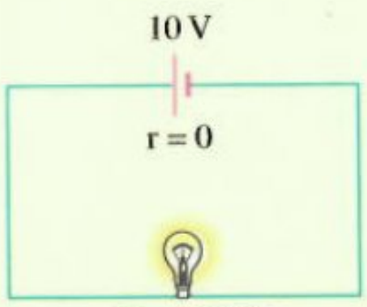




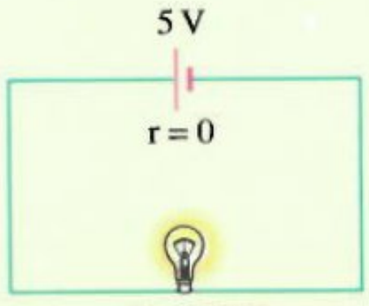
(a)



(b)



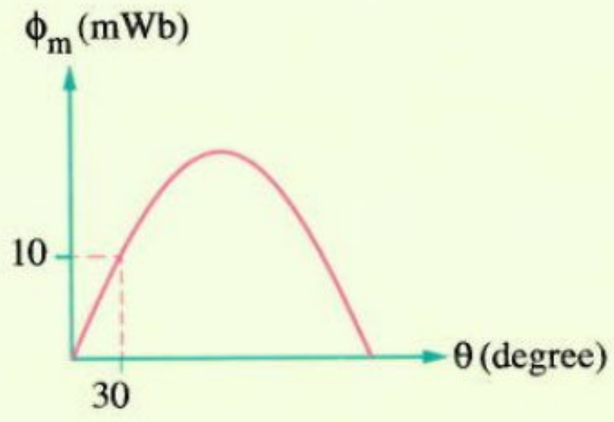
(c)



(d)

(1st session 24)

26 The opposite graph represents the relation between the magnetic flux (ϕ_m) which penetrates the dynamo coil and the rotation angle (θ) starting from the parallel position to the magnetic field lines. If the number of turns of the coil is 50 turns and rotates with a frequency 50 Hz, the maximum induced emf in the dynamo coil equals ..... . ($\pi = 3.14$)



(a) 222.2 V (b) 314 V (c) 30.8 V (d) 200 V *(2nd session 24)*

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27 The current passing through the coil of a unidirectional current dynamo ..... . *(Practice 25)*

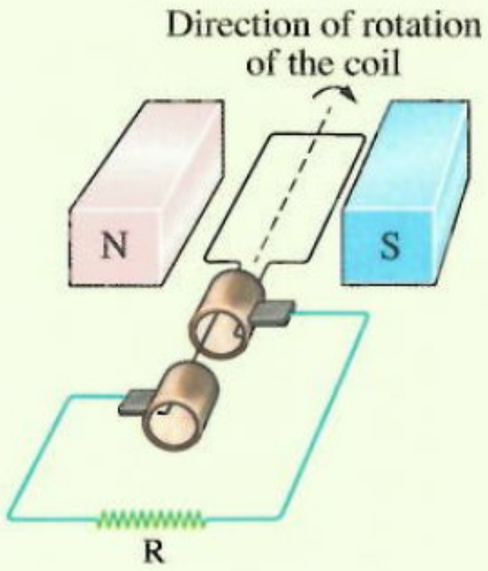
(a) changes its direction every cycle

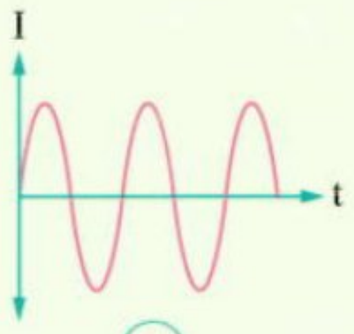
(b) changes its direction every half cycle

(c) changes its direction every quarter of a cycle

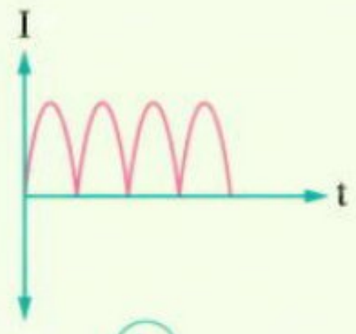
(d) always in the same direction

28 The current which is generated from the opposite device and passing through the resistor (R) is represented by the graph ..... . *(Practice 25)*

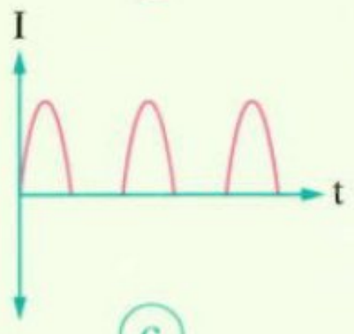




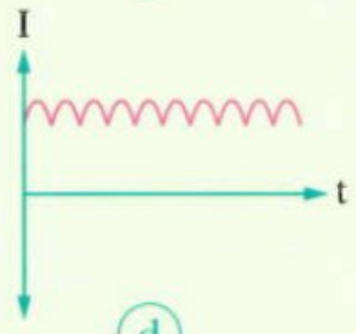
(a)



(b)



(c)



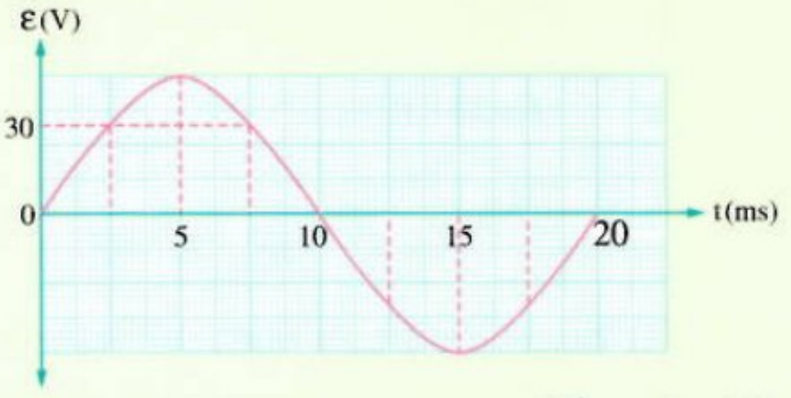
(d)

29 An AC dynamo, its instantaneous indu ced emf is calculated from the relation; $V = 200 \sin (18000\ t)$. If its coil starts rotating from the perpendicular position, the required time to reach (−100 V) for the first time is ..... . *(Practice 25)*

(a) $\frac{1}{600}$ s (b) $\frac{5}{600}$ s (c) $\frac{1}{100}$ s (d) $\frac{7}{600}$ s

30 The graph shows the change in the induced electromotive force (ℰ) in an AC dynamo coil of negligible ohmic resistance, when it rotates at a uniform angular velocity (ω).

If the angular velocity of the dynamo coil rotation increases to (2 ω), then ..... .

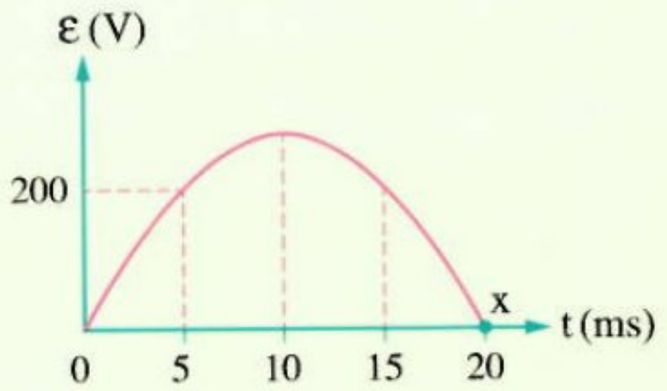


(1st session 25)

	Periodic time of coil rotation	Effective value of electromotive force
(a)	40 ms	60 V
(b)	40 ms	15 V
(c)	10 ms	60 V
(d)	10 ms	15 V

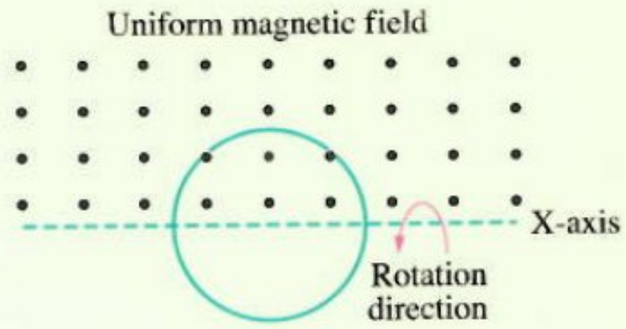
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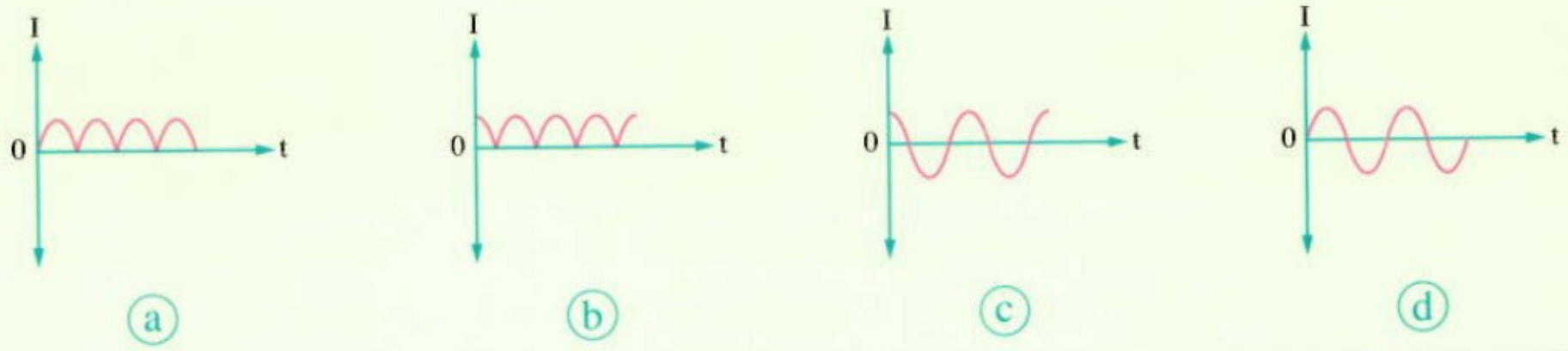
31 ✳ An AC dynamo has a coil of 100 turns. The graph represents the relationship between the induced electromotive force (ε) and time (t) over a half cycle What is the value of the magnetic flux passing through the cross-section of the coil at the moment (x)? (Consider: $\pi = 3.14$) *(2nd session 25)*



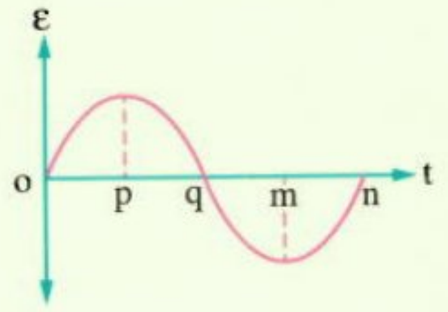
(a) 0 Wb
(b) 2.5×10^{-2} Wb
(c) 9×10^{-3} Wb
(d) 1.8×10^{-2} Wb

32 The opposite figure shows a circular coil of one turn in the plane of the page, half of the coil lying in a uniform magnetic field directed out of the page. The coil rotates at a constant speed from the vertical position about the X-axis. Which of the following graphs correctly describes the change in induced current I in the coil with time (t)? *(Practice 26)*





33 The opposite graph illustrates the relationship between the instantaneous induced electromotive force (ε) generated in a coil of an alternating current dynamo during a complete cycle. During which time interval is the average induced emf of the rotating dynamo coil equal in magnitude to its value during the interval (op)? *(Practice 26)*



(a) During interval (pm).
(b) During interval (om).
(c) During interval (on).
(d) During interval (qn).

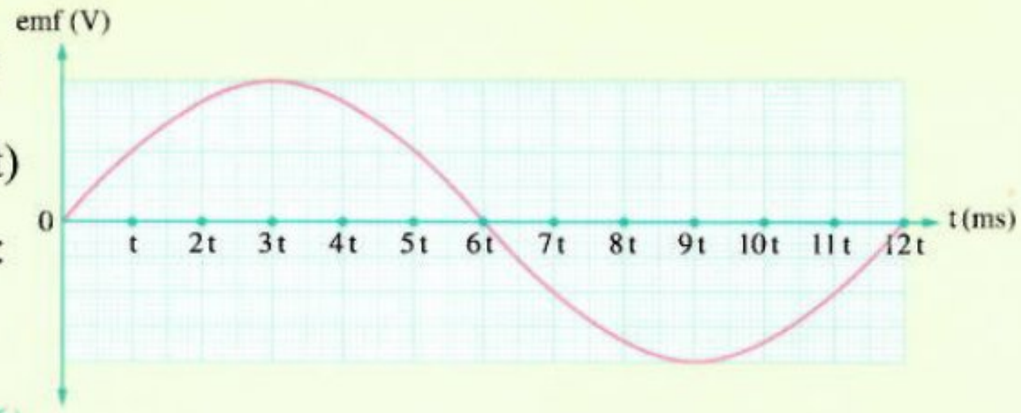
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34 The opposite graph shows the relation between the instantaneous induced emf in the coil of an AC dynamo and time (t) through one cycle, so the ratio between:

$$\frac{\text{emf during the interval from 2t to 7t}}{\text{emf during the interval from 5t to 10t}}$$

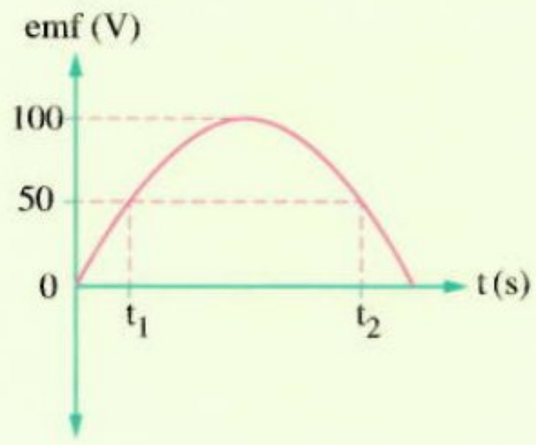
equals ..... . *(Practice 26)*



(a) $\frac{2}{5}$ (b) $\frac{2}{3}$

(c) $\frac{1}{1}$ (d) $\frac{1}{5}$

35 The graph shows the relation between the instantaneous induced electromotive force (emf) generated in AC dynamo coil and time (t). The average induced electromotive force in the coil during the time interval from (t_1) to (t_2) is equal to ..... . *(Practice 26)*



(a) 82.7 V (b) 87 V

(c) 90 V (d) 95.5 V

36 In an experiment for making a simple electric motor, a wire of length (ℓ) is wound in the form of a square coil of one turn which is rotated by an angular speed (ω) between the poles of a magnet, so the maximum induced emf in the coil was equal to (V). The coil is replaced with another where a wire of length (2ℓ) is wound in the form of a square coil of one turn also, and then rotated in the same angular speed (ω) between the poles of the same magnet, **calculate** the maximum induced emf in the coil in terms of (V). *(Practice 26)*

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